



Local Language Models for Spoken Rover Task Planning with Clarification Dialogue

by

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Research assignment presented in partial fulfilment of the requirements for the degree of Bachelor of Engineering (Electrical and Electronic) in the Faculty of Engineering at Stellenbosch University

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2026

Someone.

Acknowledgements

* Supervisor X.
* Funding Y.

> *A very long quote*
> *from someone very important.*
> – Mr M

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Date: November 2026

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Abstract

Uittreksel

Table of Contents

Abstract **iii**

Uittreksel **iv**

Table of contents **v**

List of figures **vi**

List of tables **vi**

Nomenclature **viii**

Acronyms and abbreviations ix

1. Introduction **1**

1.1. Background 1

1.2. Problem Statement and Research Questions 2

1.3. Objectives 3

1.4. Scope and Limitations 4

1.5. Report Outline 5

2. Literature Study **6**

2.1. Language models as robot task planners 6

2.2. Grounding plans in a bounded action set 8

2.3. Clarification and ambiguity 10

2.4. Locally hosted models 12

2.5. Compositional depth and hallucination 13

2.6. Speech recognition for local languages 14

2.7. Research gap 15

3. System Design **16**

3.1. User and functional requirements 16

3.2. System architecture 16

3.2.1. Off-board inference and the thin-client split 16

3.2.2. Two-rate control: waypoint intents versus the reactive layer 19
3.2.3. The end-to-end pipeline 20
3.2.4. Status lights 20
3.3. The capability vocabulary 21
3.4. Clarification loop design 22
3.4.1. Where the loop sits 22
3.4.2. The dialogue as a state machine 23
3.4.3. Checking the plan before reading it back 23
3.4.4. Confirmation 24
3.4.5. The two later moments 24
3.5. Failure recovery 25
3.6. Conditional commands 26
3.7. Navigation as instrumentation 27
3.7.1. Free-roam approach to a named object 27
3.7.2. Looking around: scan and survey 29
3.7.3. The line follower as the reactive layer 30
3.8. Reporting what happened 31
3.9. Summary of design decisions 31

4. Hardware **33**
4.1. The Turbopi platform 33
4.2. The bridge the brain actually uses 34
4.3. Reusing the stock line follower 34

5. Implementation **35**
5.0.1. Speech input and output 35
5.0.2. Speech output: Piper 36
5.1. Model inference layer 37
5.1.1. Planner 37
5.1.2. Vision-language verification 38
5.2. Clarification and mission orchestration 39
5.2.1. Clarificaiton manager 39
5.2.2. The mission execution loop 40
5.2.3. Failure reovery in code 41
5.2.4. Mid-mission revision 42
5.3. Rover client and reactive layer 43
5.3.1. Real hardware: PiRoverController 43
5.3.1.1. Virtual testing: Virtual Rover 44
5.4. Communication protocol and dashboard 45
5.5. Logging and measurement instrumentation 46

6. Experimental Evaluation **48**
6.1. Experimental methodology . . . 48
6.1.1. Command corpus design . . . 48
6.1.2. Evaluation metrics . . . 49
6.2. Plan correctness under vague commands . . . 49
6.3. Executability on the physical chassis . . . 51
6.4. Compositional depth and hallucination onset . . . 52
6.5. Reference comparison against cloud models . . . 54
6.6. Discussion . . . 55

7. Conclusion **56**
7.1. Answers to the research questions . . . 56
7.2. Contributions . . . 56
7.3. Limitations . . . 56
7.4. Recommendations for future work . . . 56

References **57**

A. Appendix title goes here **59**

List of Figures

2.1. Process overview of RAP generation before rover departures 8
2.2. SayCan combining language-model relevance scores with value-function affordance scores to select the next skill, reproduced from Ahn et al. [1]. 9
2.3. The KnowNo pipeline: an LLM planner proposes next steps with confidence scores, conformal prediction narrows these to a calibrated prediction set, and the rover asks for help whenever that set contains more than one option. Reproduced from Ren et al. [3]. 11
2.4. The introspective planning pipeline: a knowledge base of explained past examples is built offline, and at deployment the most similar example is retrieved to inform the LLM's confidence for each candidate step. Reproduced from Liang et al. [4]. 13

3.1. The compute split. The three models and the dialogue run on the brain. The Pi runs only what must keep working without them. 18
3.2. One command from speech to report. Shaded blocks call a model. The dashed block drives the rover. Dashed arrows show the scene digest feeding the two checks that need to know what is in the room. 20
3.3. The dialogue phases up to the start of a mission. A rejection returns to `CLARIFYING` with the person's reason added to the history. 24
3.4. The recovery loop. The turn counter is shared across the whole mission, so it cannot be used to drag a failing mission on. 25
3.5. How a condition is resolved. An answer the camera cannot support does not pick a branch. It stops the step and goes to the recovery loop. 27
3.6. The free-roam approach. Each cycle is one hop and one look. Every exit that is not `ok` returns `blocked` with a reason, and that is what starts recovery. . . 28

4.1. What the brain actually reaches on the Pi. Everything above `rosbridge_websocket` and `web_video_server` is Hiwonder's stock ROS2 stack, started as one unit by `bringup.launch.py` inside the Pi's Docker container. 34

5.1. `transcribe_audio`. The hallucination check exists because `fluister-turbo` occasionally transcribes silence or background noise as one of a handful of fixed captioning artefacts, most often "thank you so much for watching". . . 36
5.2. Synthesis runs on a worker thread because `_speak` shares an event loop with `ingest_frame`: synthesis is sub-second but still blocking, and perception should not stall mid-mission for a voice line. A failure degrades the mission to text-only rather than stopping it, the same soft-fail stance the pre-departure dialogue's own `_speak` in `pipeline.py` takes. 36
5.3. Both roles call the same interface. `get_provider(role)` resolves `PLANNER_PROVIDER` or `VLM_PROVIDER` to a concrete class and caches the instance. 37
5.4. `generate_plan`. A smaller model sometimes writes a step as a bare string instead of an object, or nests the list oddly, so every step is checked before anything downstream calls `.get` on it. `_promote_moves_to_approach` rewrites a `move` step whose target names an object into an `approach` step (Section 3.3); the retried transient failures from `provider.complete()` are omitted here for space and described in the text above. 38
5.5. `_ask`. Salvage is only enabled for `inventory_frame` and `survey_frame`, the two calls that ask for a list of objects and sometimes hit `num_predict` mid-list on a model that has started repeating itself; a reply with ten correct objects and one truncated one is more useful than discarding it. Every other call keeps `salvage=False`, since a truncated yes/no answer or bounding box is not something a partial parse can safely recover. 39
5.6. `_advance_to_confirmation`, one turn of the clarification loop. The repeat-question check exists because a model that cannot think of anything new to ask would otherwise repeat the same question until the five-turn cap was reached, rather than giving up as soon as it had nothing left to add. 40
5.7. `run_mission`'s step loop. The conditional-tag check is what Section 3.6 relies on: the planner only decides which steps carry a `when` tag, and this check is what honours it. The loop's own `finally` block (not shown) writes the report for every non-aborted outcome; an operator-triggered `asyncio.CancelledError` skips straight to `rover.halt()` instead. 41
5.8. `_guidance_loop`. A "no" out of the turn-count check on the right loops back to ask again rather than ending the mission; running out of turns is itself a third halt reason (`guidance_exhausted`), alongside timeout and decline, so a batch of failed runs can be told apart by why they failed. 42

5.9. `ingest_frame`'s three tiers, feeding `revision.classify`. Tier 0 is a 32×32 greyscale thumbnail diff plus a Laplacian-variance blur score (`PERCEPTION_MIN_SHARPNESS`, `PERCEPTION_MIN_INTERVAL_S`, `PERCEPTION_CHANGE_THRESHOLD`); tier 1 is one `describe_frame` caption; tier 2 only fires while executing. 43

5.10. `_follow_line`, the supervisor added around the stock line-follower node. It owns exactly the three things the node has no opinion about: arrival, obstacle stopping through the polled sonar (using `_sonar_seq` to tell two fresh readings from one stale one read twice), and a wall-clock give-up, since the node itself drives forward indefinitely if the line is lost. The `_lf_unavailable` check (added 11 October) means a drive that never started is reported as blocked rather than as a drive that ran and found nothing. 44

5.11. `VirtualRover.execute_step`'s dispatch. An `approach` step can only fail by planning (no path, or an unresolved target logged as a hallucination for RQ2), never by collision; a `move` or `turn` step is swept against the same grid and stops at the last free point if blocked, so a collision there means the chain of plan steps itself went somewhere it should not have. 45

5.12. `/ws/dialogue` (top pair) carries everything up to confirmation; `/ws/execution` (bottom pair) carries the mission. One frame per complete utterance, not a stream. 46

5.13. `report.py`'s outcome classification. A mission that needed at least one guidance turn is never reported as a plain `completed`, even if it finished, because the distinction between a clean run and a recovered one (Section 3.5) is exactly what RQ1 needs to see; `recovered` only counts a run that actually reached its target afterwards, not merely one that ended without halting. . . 47

6.1. (a) Plans with the correct visit order out of three runs, against chain depth, at three context settings (points are offset horizontally for legibility). (b) Projected tokens per planning call against depth, measured to depth 30, with the 1024 and 2048 limits marked. 54

List of Tables

1.1. Levels of command complexity considered in this project. 3

3.1. Functional (FR) and non-functional (NR) requirements, with the objective each one serves. 17
3.2. What the design provides for each command level. 18
3.3. The eight light states. 21
3.4. A conditional plan, as the prompt shows it to the planner. 26
3.5. The main design decisions. 32

6.1. Physical rover outcomes per command level, final batch (Gemma 4 E4B for planning and vision). 50
6.2. Sonar reading after the rover stopped, per case (mm). Threshold 300 mm. . . 51
6.3. Vision-model distance estimates against measured distance (visible frames only). MAE is the mean absolute error; the rank correlation is not defined for Qwen in bucket mode because its output was constant. 52
6.4. Virtual chains by kind and depth. Each cell gives correct plans and completed missions out of the runs ("Corr., compl."), and the median planner latency in seconds ("Plan"). 53
6.5. Correct plans out of three runs, by context setting and chain depth (Gemma 4 E4B). 53

Nomenclature

Variables and Functions

Constants

c	A constant.

Functions

f	A function.

Variables

$\mathbf{x}$	A variable.

Acronyms and Abbreviations

API	
CPU	
FPS	
HRI	Human-Robot Interaction
IMU	
LLM	Large Language Model
SLAM	
STT	Speech-to-Text
VLM	Vision-Language Model

Definitions

S

Student is an entity needing a thesis to transcend the state of being a student.

Chapter 1

Introduction

1.1 Background

A robot that takes spoken instructions removes two obstacles that make most mobile robots difficult to operate day to day. Typically a mobile robot operates using a screen to navigate, and an operator trained on its command set. This matters even more for a small indoor rover, because the tasks it can be asked to do, such as checking whether a door is open or looking at what is on a shelf are exactly the kind of tasks a person would rather say out loud than type into an application.

Large language models (LLMs) are neural networks which are based on the transformer architecture, and are trained on very large collections of text. Through this training the LLM's learns to generate the next word in the sequence of text given the previous words in the sequence. This makes LLM's well suited to interpreting everyday instructions as well as have a huge general knowledge base of the world. For example, when given a natural-language command and a description of what a robot can do, an LLM can break the command into a sequence of simple actions called a (RAP). This RAP can be used by any robot, and can be made up of actions that the robot is able to perform, without a developer having to hand-code a fixed list of recognised phrases [1]. Ambiguous commands can also be resolved through clarifying questions, when a command leaves out information, the model can ask the user follow-up questions before it commits to a plan. This exchange, in which the user and the robot take turns to communicate until the task is clear, is referred to in this report as a *multi-turn dialogue* [2].

Handing the planning to a language model introduces two problems. The first problem is that spoken commands are often ambiguous. So systems that always acts on its first interpretation will act on the wrong one often enough to matter [3]. The second problem is that language models hallucinate. They produce fluent, confident output that does not correspond to anything real, and this becomes more likely as a task is broken into more steps [4]. The techniques that detect ambiguity or hallucination before a plan is set

in stone can also be used while the plan is being carried out, by monitoring the robot's observations for signs that something has gone wrong [5, 6].

The systems that demonstrate these techniques typically rely on large models hosted in the cloud and accessed over the internet. These systems also typically run on robots with rich sensing such as , an inertial measurement unit (IMU) and wheel encoders that measure how far each wheel has turned. While both assumptions are reasonable in a research laboratory, they do not hold in every setting in which a voice-commanded robot would be useful.

A use case of running local only models on a robot is in a private home. Every command sent to a cloud-hosted model, and every camera frame used to check the robot's progress. These commands and images leaves the home and is processed on a third party's servers, which many households would not accept for a device that moves through their living space and records it. A home robot also has to be affordable, which rules out and a powerful on-board computer and leaves a camera as the most practical source of information about its surroundings. Low-cost drive platforms often also lack wheel encoders and an IMU, and their wheels slip enough that a command such as "drive forward one metre" cannot be trusted to be carried out accurately. In this setting, the language models must run on hardware the household already owns, such as a desktop computer with a consumer graphics card, and the robot has to rely on its camera to know whether its plan is working. Where privacy is not a concern and a reliable internet connection is available, a cloud-hosted model remains the better choice, since it is larger and more capable than any model that fits on consumer hardware.

1.2 Problem Statement and Research Questions

There is not a full understanding of whether language and vision models small enough to run on a single consumer desktop computer and can reliably turn a vague spoken command into a plan that a low-cost indoor rover can carry out using only its camera and simple on-board sensors. It is also not known how much complexity such a command can contain before the locally hosted models begin to produce plans that do not match the command, the room or the rover's capabilities. Existing work either assumes large, cloud-hosted models, or does not connect a spoken clarification dialogue to a physical robot that must wait for the user's confirmation before it moves.

In my skripsie, consumer hardware refers to a gaming desktop computer with an AMD Ryzen 5 3600 processor, 16 GB of system memory and an NVIDIA GeForce GTX TITAN X graphics card with 12 GB of video memory, representative of a mid-range gaming computer several years old. All locally hosted models, for speech recognition, planning and vision. This all must run within the memory of this machine.

To make these questions measurable, the commands the rover must handle are grouped

into the levels of increasing complexity listed in Table 1.1.

Table 1.1: Levels of command complexity considered in this project.

Level	Type	Example
L1	Single action	"Turn left."
L2	Sequence of actions	"Drive forward, turn left, then drive forward again."
L3	Perception-dependent	"Drive forward until you see the door."
L4	Conditional	"Check whether the door is open. If it is, go through it; otherwise, come back."
L5	Vague	"Go and see who is in the kitchen."

RQ1 Can locally hosted models, running on the consumer hardware defined above, turn a vague spoken command into a plan that the rover can execute, and which levels of command in Table 1.1 can they handle reliably?

RQ2 For each level of command that the models can handle, how many steps can be chained into a single command before the plan starts to include hallucinations, meaning actions, objects or locations that are not supported by the command, the room or the rover's capabilities?

In both questions, the rover's camera is its main source of feedback about the world. A vision-language model (VLM) uses camera images to check the plan against the room before the rover departs, and to confirm the rover's progress while the plan is carried out.

1.3 Objectives

The main objective of this project is to build and evaluate a fully local pipeline that turns a spoken, potentially ambiguous command into a plan an indoor rover can carry out, using models small enough to run on a single consumer-grade machine rather than a cloud API. To reach this, the project must:

1. Assemble a rover from off-the-shelf components on which the rest of the system can be tested, with a camera as its main sensor and an ultrasonic distance sensor and line sensor in supporting roles.

2. Determine which speech-to-text models can transcribe spoken commands accurately and quickly enough when run on the consumer hardware, including for speakers of South African English and Afrikaans.

3. Implement the rover's capabilities as a set of modular functions, and determine which levels of command in Table 1.1 a locally hosted planner can reliably map onto these functions.

4. Implement a multi-turn dialogue that resolves ambiguity in a command before a plan is generated, followed by a spoken confirmation step in which the rover reads its plan back to the user and only starts moving once the user agrees.

5. Ground each plan in the actual room, using a walkthrough scan recorded beforehand and camera frames checked during execution.

6. Measure and minimise the time taken from the end of a spoken command to a confirmed plan, without reducing the correctness of the plan.

7. Evaluate how the correctness and executability of plans change as commands move up the levels in Table 1.1 (RQ1).

8. Evaluate how the number of steps chained in a command relates to the rate of hallucination (RQ2).

9. Compare the locally hosted pipeline against equivalent cloud-hosted models in terms of plan correctness and response time.

1.4 Scope and Limitations

Since this project is concerned with turning vague spoken commands into robot action plans. Its scope is limited to the pipeline from speech to executed plan. This includes the transcription of speech, the clarification of ambiguous commands through spoken dialogue, plan generation and verification, and the spoken confirmation step. Navigation is used only as a means of carrying out and testing the plans, and improving navigation itself is not an aim of this project.

Before the rover is used in a room, the room is recorded once in a short walkthrough video. A VLM describes the key frames of this video in text, producing a written summary of the objects in the room and their layout. This summary is given to the planner, so that plans refer only to things that are actually present. Selected frames are also stored with their descriptions as a reference set, against which the rover's live camera frames are compared during a mission to check that it is where the plan expects it to be. The scan is not turned into a three-dimensional model or map, and building a general-purpose mapping system is outside the scope of this project.

Cloud-hosted models are used only as a benchmark for the locally hosted pipeline, and are not part of the system under test.

The rover is deliberately limited to a camera, an ultrasonic distance sensor and a line sensor, with no wheel encoders, IMU or . This reflects the low-cost home robot described in Section 1.1, and forces the pipeline to rely on the camera to determine whether a plan is succeeding, which is the capability this project sets out to test. As a result, movement is carried out without precise measurement of distance or heading, and commands that require precise distances beyond what the ultrasonic sensor can measure are not supported.

Similarly, the system is not designed to operate in arbitrary rooms or with arbitrary commands. Evaluation is limited to a small number of test rooms and a fixed set of test commands at each level in Table 1.1, which is sufficient to answer RQ1 and RQ2. The results therefore describe the pipeline's behaviour on the cases tested. Rather than guaranteeing its performance in other rooms or with other commands.

1.5 Report Outline

The remainder of this report is structured as follows: a review of related work in task planning, ambiguity resolution, and hallucination in robotic systems; the design of the speech-to-plan pipeline and its clarification loop; the rover platform and the interface between the planning software and the hardware it controls; the test methodology, including the simulated executor used in place of physical hardware; the results and discussion for each research question; and a concluding summary with recommendations for future work.

Chapter 2

Literature Study

2.1 Language Models as Robot Task Planners

Large language models carry a great deal of general knowledge about the world, but that knowledge is not grounded in what any particular robot can actually do. An LLM asked to "tidy the kitchen" has plenty to say about what tidying a kitchen involves, and none of it accounts for whether the robot in front of it has an arm, what that arm can reach, or whether "the sponge" refers to something the robot has ever seen. To Close the gap, connecting the model's reasoning to a specific robot's actual capabilities, is what the systems in this section have in common, even though they close it in different ways.

SayCan [1] grounds the model by constraining its output rather than trusting it outright. The LLM proposes a set of candidate next actions drawn from a fixed library of pretrained skills, but each candidate is re-scored by a separately learned affordance function. that estimates how likely that skill is to succeed given the robot's current state. The action that is both something the LLM would suggest and something the robot can plausibly do is the one that gets executed. Grounding, in this framing, is a second model checking the first, not something the LLM is trusted to know on its own.

Code as Policies [7] takes a different route. Rather than picking from a fixed skill library, the LLM writes code, calling perception functions, control primitives, and other functions it has been shown examples of, composed however the instruction requires. This lets it handle instructions no one anticipated in advance and assign precise parameters (a velocity, a threshold) to descriptions that were only ever given in words, at the cost of needing a language model capable of writing correct, composable code rather than just picking from a short list.

Inner Monologue [8] addresses a limitation both of the above still have: a plan committed to once, before anything happens, has no way to adapt as the situation unfolds. It closes the loop by feeding the robot's ongoing observations, success and failure signals, scene descriptions, even a human's spoken interjection, back into the language model's own

reasoning process, so it can revise what it does next in light of what actually happened rather than what it expected to happen.

None of these three were built for tasks longer than about ten actions. Hori et al. [2] target that gap directly, arguing that long-horizon tasks fail for a different reason than short ones: the command usually leaves out information the model would need to plan the whole thing correctly, and no amount of re-scoring or code generation fixes a plan built on a wrong assumption. Their method resolves this through what they call active modification, the model asks clarifying questions before committing to a plan, combined with passive modification, refining the plan afterwards against visual feedback. This is the most direct ancestor of the clarification loop this project builds: extending plan length by first closing the information gap in the command, rather than by making the underlying model larger or its skill library richer.

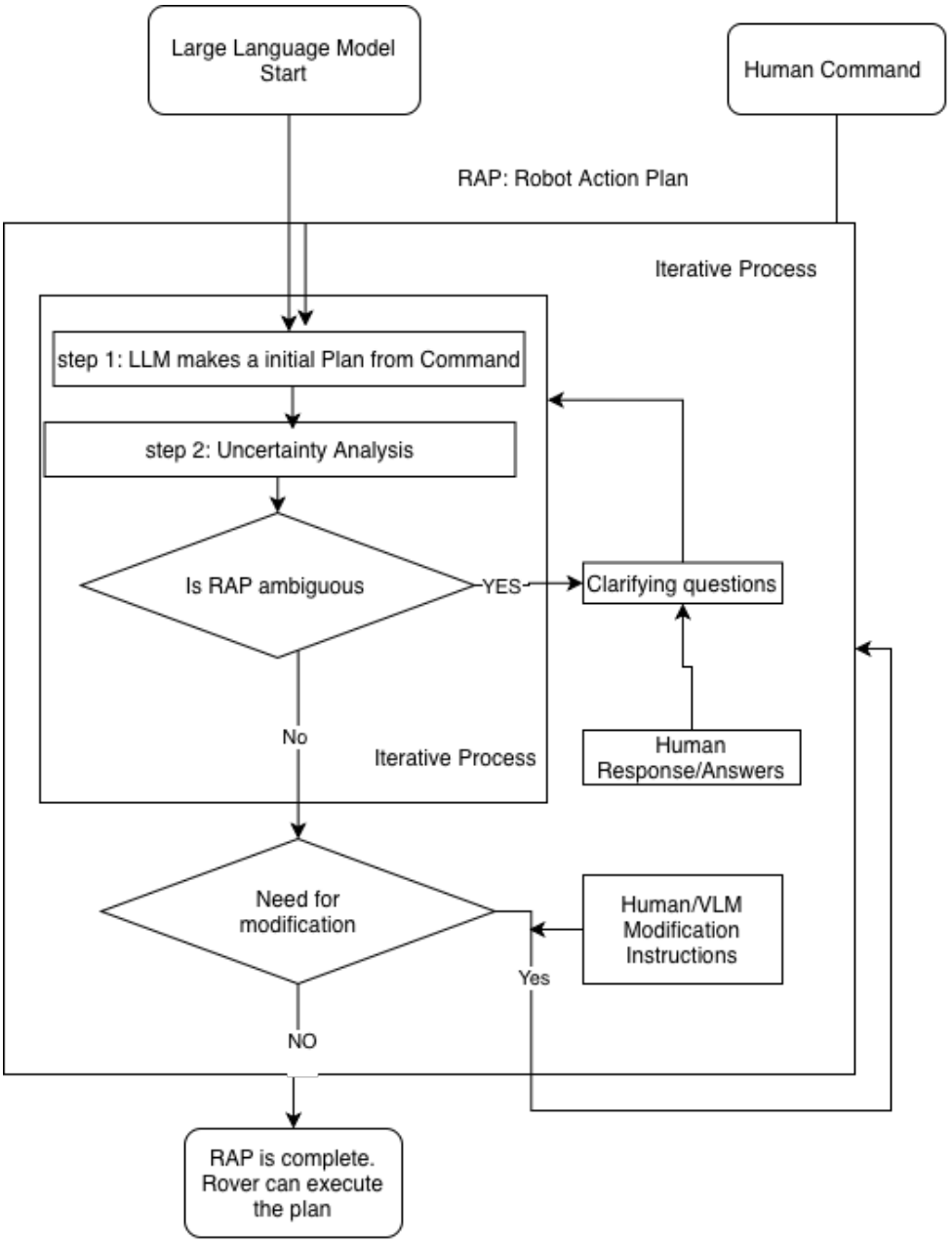


Figure 2.1: Process overview of RAP generation before rover departures

Last point Code as Policies' approach assumes a language model capable of writing correct, composable code against a rich API, which is a reasonable assumption for a cloud-hosted model that has more computing power but not for the small, locally hosted planner this project depends on for RQ1.

2.2 Grounding Plans in a Bounded Action Set

SayCan and Code as Policies set the standard way to connect an LLM's plan to what a robot can actually do. SayCan [1] pairs the LLM with a separately learned value function for each robot skill. The language model scores how relevant an action sounds for

the instruction. The value function scores whether the robot can do it right now. The two scores are combined, so the action picked is both sensible and physically possible. Figure 2.2 shows this with a concrete instruction, "how would you put an apple on the table?" The LLM alone ranks *find an apple*, *find a coke*, and *find a sponge* almost the same, since all three are plausible first steps for some task. The value function breaks that tie: *find an apple* and the other two score identically on the language side, but only the apple is actually visible in the robot's current scene, so the combined score, log-probability from the LLM multiplied against the affordance probability, pushes it to the top of the list. Nothing about the instruction told the model an apple was present. The value function is what supplied that.

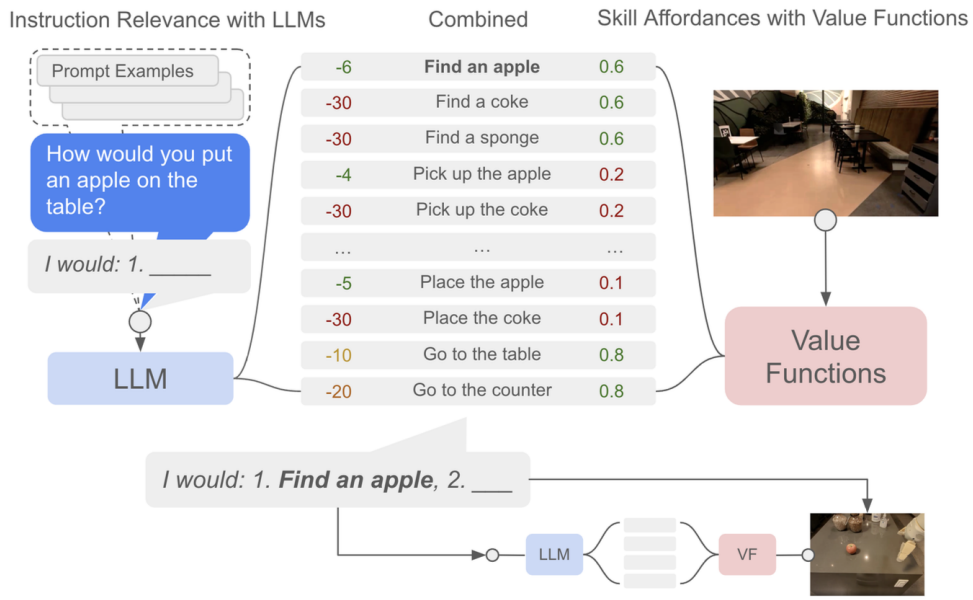


Figure 2.2: SayCan combining language-model relevance scores with value-function affordance scores to select the next skill, reproduced from Ahn et al. [1].

Ahn et al. formalise this selection rule as

$$\arg\max_{\pi \in \Pi} \; p(c_\pi \mid s, \ell_\pi)\, p(\ell_\pi \mid i) \tag{2.1}$$

where: Π is the set of available skill

ℓ_π is skill π's textual label,

i is the instruction,

s is the current state,

c_π is the event that π completes successfully.

$p(\ell_\pi \mid i)$ is task-grounding, the language model's estimate of how well a skill matches the instruction, and $p(c_\pi \mid s, \ell_\pi)$ is world-grounding, the value function's estimate of whether that skill would succeed from the current state.

These are exactly the two columns in Figure 2.2, the LLM's log-probabilities are $\log p(\ell_\pi \mid i)$, and the 0.6/0.2/0.1/0.8 values are $p(c_\pi \mid s, \ell_\pi)$.

Code as Policies [7] solves the same problem a different way: the LLM writes Python that calls a fixed set of perception and control functions directly. If the code does not compile against that API, the step cannot run.

Both approaches give the planner a fixed set of actions it cannot go outside of. This project's capability vocabulary in `planner.py` plays the same role: a step whose target or action falls outside that vocabulary is rejected rather than passed through to the rover.

Neither approach checks what happens after a step runs. Inner Monologue [8] adds that check on top of a SayCan-style setup: after each action, the robot gets an observation back, a success or failure signal, a scene description, or something the human said, and the next step is planned using that observation instead of the original plan alone. Plan a step, act, observe, replan. This project's mid-mission execution follows the same loop: each frame sent over `/ws/execution` is what the plan gets checked against after each step, not only before the mission starts, allowing the plan to respond to drift the rover has actually accumulated rather than committing to a path decided before departure.

2.3 Clarification and ambiguity in human-robot interaction

A spoken instruction only resolves to a single rover action if the rover's observations rule out every other reading of it. "Bring the bowl" is unambiguous with one bowl on the counter and ambiguous with two. Hori et al. [2] describe two ways to correct a task plan once this kind of ambiguity shows up: passive modification, where a person reviews the plan and points out what is wrong, and active modification, where the planner judges its own uncertainty and asks a question before acting.

Passive modification is accurate, since a human checks every step, but it is repetitive. The plan has to be re-issued and re-checked each time something is off. Active modification moves that judgment onto the model, so the human is only asked to step in when the model is actually unsure.

Ren et al. [3] give active modification a statistical guarantee, in a method called KnowNo, and Hori et al. cite it directly as a working example of the approach. KnowNo does not let the planner commit to its single highest-scoring next step. The planner's raw confidence scores for each candidate step are passed through a conformal prediction layer, which converts them into a calibrated set of steps that are all plausible enough to keep. A singleton set means the rover proceeds on its own. A set with more than one step means the instruction was ambiguous given what the rover can observe, and the rover asks a clarifying question instead of guessing. Figure 2.3 shows the example used in the paper.

It is told to place the bowl in the microwave with a metal and a plastic bowl both on the counter, the two matching steps score close enough that neither clears the calibrated threshold on its own, so the system asks which one, plastic or metal, instead of picking one.

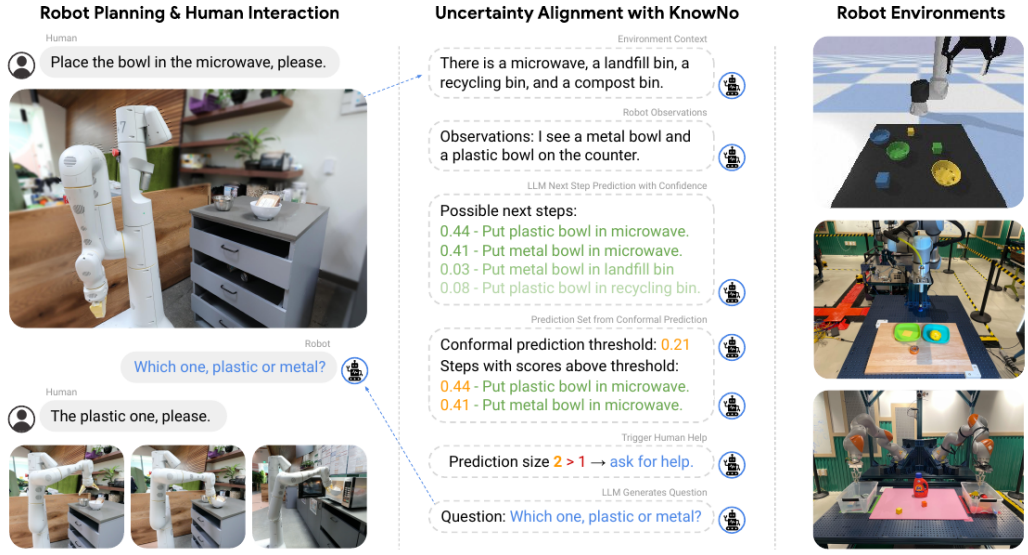


Figure 2.3: The KnowNo pipeline: an LLM planner proposes next steps with confidence scores, conformal prediction narrows these to a calibrated prediction set, and the rover asks for help whenever that set contains more than one option. Reproduced from Ren et al. [3].

The calibration works as follows. Given a held-out calibration set of N examples, each with a true label y_i and a model score $\hat{f}(\tilde{x}_i)_{y_i}$ for that label, a nonconformity score is computed for every example:

$$\kappa_i = 1 - \hat{f}(\tilde{x}_i)_{y_i}. \tag{2.2}$$

The calibrated threshold $\hat{q}$ is then set to the

$$\frac{\lceil (N+1)(1-\epsilon) \rceil}{N} \text{ empirical quantile of } \kappa_1, \ldots, \kappa_N, \tag{2.3}$$

where ϵ is a user-chosen error tolerance. At test time, the prediction set kept for a new instruction is every label whose score clears this threshold:

$$C(\tilde{x}_{\text{test}}) = \left\{ y \in Y \;\middle|\; \hat{f}(\tilde{x}_{\text{test}})_y \geq 1 - \hat{q} \right\}. \tag{2.4}$$

Conformal prediction guarantees that this set contains the correct label with the user-specified probability,

$$P\Big(y_{\text{test}} \in C(\tilde{x}_{\text{test}})\Big) \geq 1 - \epsilon. \tag{2.5}$$

This guarantee is what makes the threshold usable in practice: instead of picking a confidence cut-off by hand and hoping it behaves sensibly, the rover's decision to act alone comes with a stated bound on how often that decision is correct.

2.4 Locally hosted and quantised models

Sections 2.1 to 2.3 assume the rover can call out to a large, cloud-hosted LLM. A rover cannot always guarantee that access: network connectivity is not reliable in every test environment, and the round-trip latency to a cloud endpoint works against the response time a voice-commanded system needs to feel usable. Running the planner locally, on hardware the rover carries itself, removes both problems, but restricts the model to whatever fits in on-board memory.

Quantisation is the standard technique for making a model fit that constraint: model weights, and in some cases activations, are stored using fewer bits than the 16 or 32 bits used at training time, at some cost to numerical precision. Liu et al. [9] define uniform quantisation of a tensor $\mathbf{X}$ to b bits as

$$\hat{\mathbf{X}} = \mathcal{Q}(\mathbf{X}; b) = s \cdot \Pi_{\Omega(b)}\left(\frac{\mathbf{X}}{s}\right), \qquad s = \frac{\max(\mathbf{X}) - \min(\mathbf{X})}{2^b - 1}, \tag{2.6}$$

where

* $\Omega(b) = \{0, 1, \ldots, 2^b - 1\}$ is the set of representable integer levels
* $\Pi_{\Omega(b)}(\cdot)$ rounds to the nearest one.

This is applied at different bit-widths for weights and activations, written for example as W8A8 (8-bit weights, 8-bit activations) or W4A16 (4-bit weights, 16-bit activations); A lower bit-width gives a smaller model and a coarser step size s, and therefore more rounding error per value.

Husom et al. [10] benchmark quantised LLMs directly on edge hardware and report that quantisation lowers energy use and inference latency substantially, at a small cost to output accuracy on straightforward tasks. Liu et al. [9] show that this cost is not constant across task types. Accuracy degrades faster as bit-width drops on tasks that require chaining several reasoning steps together than on single-step tasks. The planning step described in Section 2.1 is exactly that kind of multi-step reasoning task, so the bit-width chosen for a locally hosted planner in this project is not only an efficiency setting. It is a direct trade-off against how reliably the planner reasons through a multi-step voice command.

2.5 Compositional depth and hallucination

The methods discussed in Sections 2.1 to 2.3 treat each planning step largely on its own. As a task plan grows longer and more compositional, two further problems appear that a single-step clarification mechanism does not cover: the LLM's reasoning can degrade over a long chain of steps, producing a confidently stated plan that is simply wrong, and the rover can encounter a scene it was never prepared for, one that no single faulty component causes but that the plan as a whole fails to account for.

Liang et al. [4] address the first problem by extending the conformal prediction procedure from Section 2.3 with a retrieval step. Rather than scoring each candidate plan from the LLM's raw confidence alone, their method, introspective planning, first builds a knowledge base of worked examples. For a given task and observation, an LLM is prompted to explain in writing why certain candidate actions are valid and others are not. At deployment, the system retrieves the most similar past example by embedding similarity and includes it in the prompt before the LLM scores its candidates, as shown in Figure 2.4. Grounding each new decision in a similar, explained precedent sharpens the confidence scores that conformal prediction operates on, which produces the same statistically guaranteed prediction set from Equations 2.4–2.5 while asking the user fewer unnecessary questions.

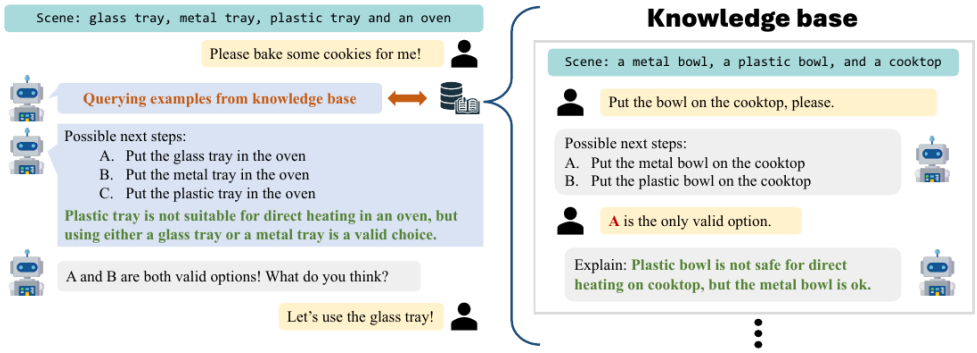


Figure 2.4: The introspective planning pipeline: a knowledge base of explained past examples is built offline, and at deployment the most similar example is retrieved to inform the LLM's confidence for each candidate step. Reproduced from Liang et al. [4].

The second problem, an unanticipated scene rather than an unreliable plan, is addressed by Sinha et al. [5] and Elhafsi et al. [6] from the direction of runtime monitoring rather than planning. Sinha et al. propose a two-stage monitor: a fast binary classifier screens every observation in an LLM embedding space, and only when it flags a possible anomaly does a slower, more capable LLM reason explicitly about what has gone wrong and which fallback plan to switch to. Splitting the monitor this way keeps the fast path cheap enough to run continuously, while reserving the expensive reasoning step for the moments it is

actually needed, which matters directly for a rover with the limited on-board compute discussed in Section 2.4. Elhafsi et al. apply the same idea of an LLM as a monitor to what they term semantic anomalies. A system-level failures, like an autonomous vehicle braking for a stop sign printed on a billboard. That does not come from any single broken component, but from a gap in the system's contextual reasoning.

Together, these three methods extend the single-step clarification of Section 2.3 along a different axis: introspective planning improves the confidence estimate feeding a single decision, while AESOP and semantic anomaly detection catch failures a confidence score never gets the chance to be wrong about, because the plan itself is being carried out against a scene the planner misjudged.

2.6 Speech recognition for South African English and Afrikaans

Every method discussed so far assumes the instruction d_i handed to the LLM planner is already correct text. For a voice-commanded rover, that text comes from a speech-to-text stage first, and an error there becomes an error in the planner's input before any clarification mechanism gets a chance to act on it. South African English and Afrikaans are both under-represented in the training data of general-purpose speech recognisers, which makes this an upstream risk specific to a South African test environment.

Kamper et al. [11] quantify how much accent variation costs a algorthmn used for recognising speech, trained without accounting for accent variation. Working with 31.78 hours of speech across five South African English accents (Afrikaans English, Black South African English, Cape Flats English, White South African English, and Indian South African English), they compare three acoustic modelling strategies: a separate model per accent, a single model pooling all accents together, and a multi-accent model that shares acoustic structure across accents where it helps while keeping accent-specific distinctions where it does not. The multi-accent model reached 82.78% word accuracy, against 81.53% for accent-specific models and 81.52% for a single pooled model, a difference the authors report as statistically significant. Black South African English stood out as the one accent that still benefited from being modelled separately rather than pooled, which the authors attribute to it being acoustically the most distinct of the five.

Jacobs et al. [12] work at the opposite end of the resource spectrum: fine-tuning Whisper on Afrikaans and isiXhosa oral narratives spoken by four- and five-year-old children, using only five minutes of transcribed in-domain child speech. They find that adding in-domain adult speech, especially combined with voice conversion to make it sound more child-like, gives the largest accuracy gain of any strategy tried, and that parameter-efficient fine-tuning helps Afrikaans but not isiXhosa, since isiXhosa is more

severely under-represented in Whisper's original training data.

Together, these results point at the same conclusion from different directions. Local accents and languages measurably degrade off-the-shelf speech recognition, and the fix in both cases is not more general-purpose training data but a small, targeted amount of in-domain or in-accent data. For a rover meant to take spoken commands from South African English or Afrikaans speakers, this is a design constraint on the speech-to-text stage that sits upstream of every planning method discussed in Sections 2.1 to 2.5, not something those methods can correct for after the fact.

2.7 Research gap

Sections 2.1 to 2.6 cover five separate strands of work: grounding an LLM's plan in what a rover can actually do (Sections 2.1-2.2), asking for clarification when an instruction is genuinely ambiguous (Section 2.3), catching the failures that a confidence score alone will not (Section 2.5), running the planner on hardware small enough for a rover to carry (Section 2.4), and recognising speech accurately in a South African accent or language (Section 2.6). Each strand has been developed and evaluated largely on its own.

The methods in Sections 2.3 and 2.5 that provide statistical guarantees on plan correctness, KnowNo and introspective planning, are demonstrated with large, cloud-hosted LLMs. Section 2.4 shows that quantisation, needed to run a planner locally, costs accuracy disproportionately on exactly the kind of multi-step reasoning these planners perform, but neither Ren et al. [3] nor Liang et al. [4] evaluate their calibration procedure on a quantised, locally hosted model. Whether a conformal prediction threshold calibrated on a full-precision model still gives a valid guarantee once that model is quantised down to fit on-board memory is not addressed by either strand of work.

Similarly, none of the planning or clarification methods in Sections 2.1 to 2.5 are evaluated with speech input, let alone speech in a South African accent or in Afrikaans. Section 2.6 shows that accent and language mismatch measurably degrades transcription accuracy, which means an instruction d_i reaching the planner in this project's setting carries a source of error the literature above does not model at all. An ambiguity introduced by a transcription error is not the same problem as an ambiguity in the underlying instruction, and it is unclear from the existing literature whether a clarification mechanism designed for the latter behaves sensibly when faced with the former.

This project sits at the intersection of these gaps: a voice-commanded indoor rover that must plan and clarify using a locally hosted, quantised LLM, taking speech input from South African English or Afrikaans speakers. No single work reviewed in this chapter evaluates this combination, and the remainder of this report investigates how the planning, clarification, and speech recognition components described above hold up once combined on hardware a rover runs on.

Chapter 3

System Design

3.1 User and Functional Requirements

The end user in mind is someone who is not staring at a screen. It is someone who would be able tel a indoor rover what to do using speech. They are also not necessarily going to watch a screen to see if the rover is on track. They can say things like 'Go look at the box on the room' or 'what is on the floor under the bed'. The rover will have to clarify if there is multiple boxes or where the bed is in the room. Ask any questions to resolve ambiguity and then go and execute it's plan.

In the table below is the requirements in Table 3.1. Each one traces back to an objective in Sections 1.3.

Since there is no odometry sensors on the provided rover. Every distance covered by the rover is a open-loop estimate. The camera and sonar sensors are used to correct this estimate. The rover travels and moves in open-loop steps but gets corrected once a step is completed or a obstacle is encountered by camera and ultrasonic sensors.

To Test how well the plans are generated. I made a table with command levels in Table 1.1 each need something different fromt he design, and Table 3.2 says what. These mappings were made as the project progressed and the design evolved. For example, at first I did not think it was necessary but it evolved as the project got more scope.

3.2 System Architecture

3.2.1 Off-board Inference

The rover used is a HiWonder TurboPi with a Raspberry Pi 5 and about 4GB of RAM. The Turbo Pi has and expansion module called the HiWonder Echo pro microphone and audio adapter connected via USB The planner, whisper translator module and Vision

Table 3.1: Functional (FR) and non-functional (NR) requirements, with the objective each one serves.

ID	Requirement	Objective
FR1	Take the command by voice and transcribe it locally.	2
FR2	Before planning, check the command against the room and ask the person a spoken question when the command is ambiguous, up to a fixed number of questions.	4
FR3	Produce a plan that uses only actions the rover has and objects the room contains.	3, 5
FR4	Read the plan back and move only after a spoken confirmation. A rejection must lead back to the dialogue, not end it.	4
FR5	Carry out the plan one step at a time, if a obstacle is in the rovers way it should stop without consulting a model.	1, 3
FR6	Use the camera to check progress and arrival during a step.	5
FR7	Report the outcome of the rover if it completed the steps and if it was unsure about anything.	4, 7
FR8	When a step fails, ask the person what to do next and carry on from where the rover is. This is for failure only, never for ambiguity.	4
FR9	Log every model call and decision made to help with debugging.	6, 7, 8
FR10	When a command contains a condition, look once to decide it, run only the matching steps, and ask the person when the camera cannot tell. Never guess a branch.	3, 7
FR11	When the person asks what is near a target, report what is around it and not only whether the target was reached.	3, 5
NR1	Run planning, vision and speech on consumer grade hardware, e.g. a Ryzen 5 3600 with 16 GB RAM and a 12 GB GPU. Cloud models are allowed only as a benchmark.	9
NR2	Keep the rover stoppable when the network or a model is slow or gone.	1
NR3	Use only a camera, an ultrasonic sensor and a line sensor. No encoders, IMU or lidar.	1
NR4	Let the planner or vision model be swapped by configuration, with no change to the pipeline code.	9
NR5	Show the person what state the rover is in without a screen, using onboard lights	4

Table 3.2: What the design provides for each command level.

Level	Command type	Mechanism
L1	Single action	One `move` or `turn` step, read back and confirmed.
L2	Sequence of actions	Ordered steps, each finished before the next starts.
L3	Perception-dependent	`approach` to find and reach a named object (Section 3.7.1), or `follow_line` to a visible end point.
L4	Conditional	A `check` step with tagged branches (Section 3.6).
L5	Vague	The clarification loop before departure (Section 3.4) and a survey of the surroundings when the command asks for one (Section 3.7.2).

model runs on a separate host machine. Keeping only the motion and sensing on the Pi itself. The reason for this configuration is due to size. The 3 models together take about 10GB of RAM and if that is excluding the stack required to drive the PI and control the camera. There is physically enough room to have a reasoning model on this hardware. Not yet atleast.

Running the models on a seperate machine with more hardware capability also means that I can swap a model by changing one environment variable, which the cloud comparison depends on.

I call the seperate machine hosting the models 'the brain'. It communicates to the turboPi over a rosbridge on a port that it exposes. Using a library called 'roslibpy' (Figure 3.1). This allows for commands from the brain like: The plan, the keyframes, vision results, and questions for the user to be sent to the rover. The Camera frames, sonar readings And step results come back to the brain.

since the brain hosts all the model decisions and state of the current plan. If there is a network drop the rover can still continue operating with the last known plan. It just cannot ask for clarification or get new plans.

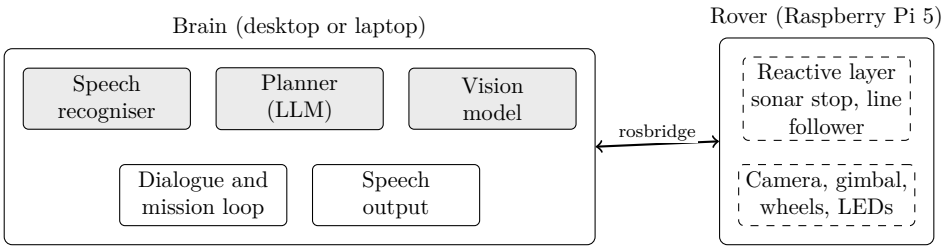


Figure 3.1: The compute split. The three models and the dialogue run on the brain. The Pi runs only what must keep working without them.

The code boundary is one method, `RoverController.execute_step`. The brain hands it one step and gets back `ok`, `blocked` or `halted`. Everything above that method is the same for the real rover and for a simulated one, so I can run the same dialogue and mission code in simulated room which is a constructed room from a JSON file before putting it on hardware.

The default configuration is fully local. Speech recognition is `faster-whisper` with the `fluister-turbo` model, the planner is `gemma4:e4b` `gemma4:e4b`, and the vision model is `qwen2.5vl:3b` or `gemma4:e4b`. Both are served by Ollama with a context of 8192 tokens.

Speech output uses Piper, a small neural text-to-speech engine that runs locally. It turns text into a waveform with a voice model stored as an ONNX file, and I use the `en_US-lessac-medium` voice. I chose it for three reasons. The rest of the pipeline is local, so the rover's voice has to be local too (NR1), and a cloud speech service would put a network call and a third party into every spoken reply. Piper also runs on the CPU so it does not compete with the Vision and planner models for VRAM. One thing to note if I use 1 model for both vision and reasoning, I would have more VRAM at my disposal.

The 8192-token window is a constraint I check instead of assuming. Every clarification turn and every recovery turn adds text to a prompt. The small local model usually silently truncate plans without raising any error. A separate offline tool, `context_budget.py`, counts the tokens each part of the prompt takes (fixed instructions, room digest, one clarification turn, one recovery turn) so that I know how many turns fit before the window is the limiting factor.

3.2.2 Two-rate Control

Everystep or run requires the rover to do a vision-model call. In order to process, where it is at the current time and if it is still on track. A vision-model call takes about 4.78 s on my hardware. If the rover is using the line follower to control the driving, The steering of the rover cannot occur in a closed loop.

Instead there is a reactive layer on the turboPi itself, that came stock with it. It reads the sonar at 10-30 Hz and stop the rover if something is in the way. It also follows a line from the IR array uder the rover without consulting the models. The actual decision making engine runs on the brain at a pace of the models and hardware. It chooses the next steps, deciddes whether the rover has arrived at the desired destination. Where the closed loop system occurs is at every step boundary and at distinctive times during the rovers runs.

The reason it is design like this is because the rover never relies on the vision model to make a immediate decision on when to stop If a obstacle arises infront of the rover.

3.2.3 Pipeline

The Figure 3.2 gives you a the overview of the process of one command, given to the rover. A short walkthrough video is recorded of the room the rover will be traversing ing. The then breaks up the video of the room into keyframes (the number depends on the environment variable set). The vision model them lists all the objects it sees in the in each frame as well as any obstructions in the image. This is called the scene digest which can be called on by the model to understand the environment. It is capped at 8 major objects with each having 2 attributes for each object. The reason it was capped at 8 objects so that it fits the planners prompt without the dialogue.

I chose text over a map delibreatly. A map is a seperate system to create and keep updating and generating one is outside the scope of my Skripsie.

A spoken command by someone is then transcribed and enters the clarification loop (Section 3.4). If the command is not flagged as ambiguous the planner can turn is into steps using the actions and vocabulary/actions the rover can execute. (Section 3.3). The vision model then checks the plan against the digest for steps that are not able to be executed. If it comes back successful the plan is read back to the command giver and the rover only moves after they confirm the plan. The rover then run the steps one at a time and at the end reports are logged and spoken.

A Websocket will be used to carry the dialog sessions. `/ws/dialogue` will serve everything up until the confirmation, and `/ws/execution` servers all the communication during the mission run. I keep them apart so that the dialogue state and the mission state never share a single session.

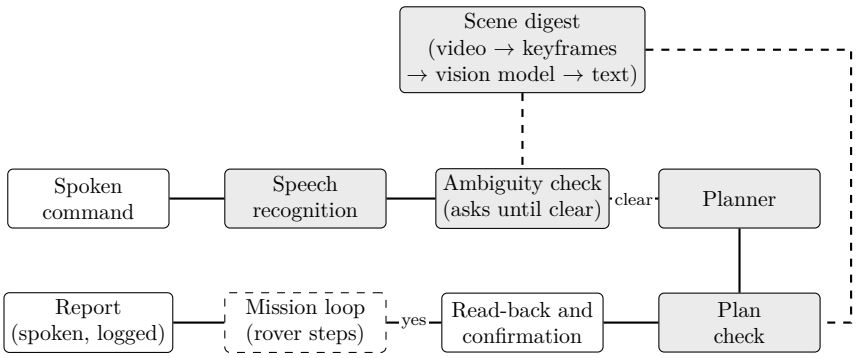


Figure 3.2: One command from speech to report. Shaded blocks call a model. The dashed block drives the rover. Dashed arrows show the scene digest feeding the two checks that need to know what is in the room.

3.2.4 Status Lights

To provide the user with some feedback on the rover state throughout the mission so that they can tell what the rover is doing. The Pi has RGB LEDs the expansion board as well

as the ultra sonic sensor ring. I used them as a state display (Table 3.3). During each phase of the mission execution as well as the dialogue session before a mission is executed. Each light color is tied to one state. If the light setting failed it should not hinder the rovers execution therefore Rovers that have no LEDs inherit a no-op, and one setting (`RGB_ENABLED`) turns the whole thing off.

Table 3.3: The eight light states.

State	Pattern	Meaning
listening	steady blue	Waiting for a command or an answer.
thinking	red and green alternating, 0.5 s	A model is working.
confirm	steady amber	A plan has been read back and the rover is waiting for yes or no.
executing	steady green	Carrying out a step.
searching	steady cyan	Looking for a target.
awaiting_guidance	magenta, 1 s on and off	Stopped and waiting for the person to say what to do next.
arrived	three green flashes, then steady green	A target has been reached.
blocked	steady red	The mission has stopped and cannot continue.

The mission logs also log the changes with timestamps, therefore allowing one to reconstruct the mission run later on or use it for debugging.

3.3 The Capability Vocabulary

Since these planners are LLMs and have generalized knowledge, a planner based on a LLM may write any step it likes and will also write steps that the rover cannot execute. Commands like 'bring me that bottle' invites a `pick_up` step. Since our rover has not arm. I gave the planner a fixed list of actions and in the prompt describe which exists and what each of these actions means.

The list depends on the rover, so it lives in a planner profile. If a different hiWonder Rover is chosen, the actions can be overridden using a different configurtion of Planner. The Real rover has the following actions: `follow_line`, `move`, `turn`, `approach`, `scan`, `observe`, `stop` and `report`. Every step that the rover executes is a small JSON object with an id, an action and a target.

The simulated rover has the same actions except for `follow_line`. The `move` and `turn` step both carry a distance and an angle. The reason for that is the simple fact that the simulated rover can measure distance travel since the virtual room in which it travels has a define width and length. The real rover does not have this information about the room in which it is traveling. This was chosen in (NR3). The simulated rover also cannot

pick up things as it has no arm and cannot carry anything. These are all decided in the planner profile for the simulated rover.

There is a ninth action that is `check`, this allows the rover to 'look at the destination object and decide whether the condition is met' (Section 3.6). This action only takes place when the command contains a words like: *if*,*whether*, *otherwise* or *unless*. The prompt then gets a extra block that explains the `check` step as well as the optional `when` option for the alternative. This gives the rover 2 options to choose from before executing the plan. The checked field is left out of everyother prompt for two reasons. The first reason is the extra line is more tokens take up in the already small context window of 8192-tokens. The second is if there is extra context that the rover does not need it can results in a plan being miss interpreted or inventing a condition that nobody spoke of. There is a limiting factor of deciding to use the `check` action with a word match is that conditional phrased in a different way e.g 'when you see the door stop', will not be translated as conditional by the planner. Conditional logic also only exists on the real rover's profile and not the simulated rover.

The first plan generated by the planner that has actions outside of the list is not rejected. There are three methods used to catch it afterwards. A step with an unknown action returns `blocked` status if the rover tries to run it. The reporting phase in the brain counts the out-of-vocabulary actions in each plan the rover executes in orer to measure how often the planner ignors the list (section 6.3). Before the rover goes off on its mission the person hears the entire plan. I chose to measure the first-plan failure instead of just reprompting it and fixing the plan. The only plan action that is edited on the fly is the `move` action when the rover is in free mode. It changes the action to `approach`, since these two actions are similar and the planner will try to use move instead of approach.

The plan generated by the planner when the rover halts and ask for further instructions. This plan is used in the mission as is and not read-back to the user. This means that every step is validated to make sure the action is in the profile of the rover. Any plan that fails this check is discarded (Section 3.5). In free-roam mode the list for a replan leaves out the `follow_line`, since there is no floor line for it to follow.

3.4 Clarification Loop Design

3.4.1 Placement

Some spoken commands are often underspecified, and if the rover executes on the wrong assumption it could place it in the wrong place. The ambiguity is handeled at 3 distinct places.

Before the departure, the system asks for a clarifying question if the command is unclear or ambiguous. During the mission it does not stop to ask about any ambiguity. It

compares what is currently can see with the digest and corrects itself as best as it can (Section 3.4.5). When it reaches its destination it communicates what is still was unsure about. Any mid-mission ambiguity questions are left out on purpose since the command speaker might not be there to answer the question but there is an exception for if the rover halted and cannot continue.

A rover that tries its best to execute a plan and get to a distinace is far better than one which stops and waits for a reprompt on a ambiguity it spotted in it's current plan. This is different to what was seen in KnowNo [3] and Introspective Planning [4], which ask for help when their confidence drops.

3.4.2 State Machine

The command spoke moves through multiple phases in Figure 3.3. During the CLARIFYING, the vision model is used to determine whether the command is ambiguous. Using the given command and the scene digest. The vision models answer is in JSON with four fields: wether the command is ambiguous, why ,a question to ask, and a rewritten command. If it is ambiguous, the quesiton is spoken and the answer is added to the history. If it is clear, the rewritten command is sued from then on. The command only names objects and avoids words like 'left' or 'behind you'. Since those are relative to the camera view at the time of asking and can be completely wrong once the rover moves.

This clarifying loop is capped at five questions. If the command is still unclear after that, the system marks the session as capped and plans from the rewritten command if the model produced one after a question was answered or from the original command if no clarifying question was answered. The loop also drops a quetion it has already asked. If that is the only question that the model produces then there is nothing new to ask and the loop breaks and treats this as a cap moving onto the planning. A warning is added to the output plan when it is read-back to the user which starts with 'I am no fully sure I understoon', so the person knows the plan is not 100% certain. There is a cap to the number of questions asked since the user can get tired of answering questions or it can result in a endless ambiguous loop of questions and answers.

If the vision model ever fails to load. It is retired once and if it fails again, the person is told and asked to repeat their command again. Not having this fall back coould result in letting every command through.

3.4.3 Plan Check

After the planner model produces a plan. The vision model can then check it against the digest and returns whether it is workable and a list of concerns. If it says no, the planner runs once more with the concerns added as advice while the original goal stays as it was. The replan happens only once. Replanning a second time rarely fixes what the first replan

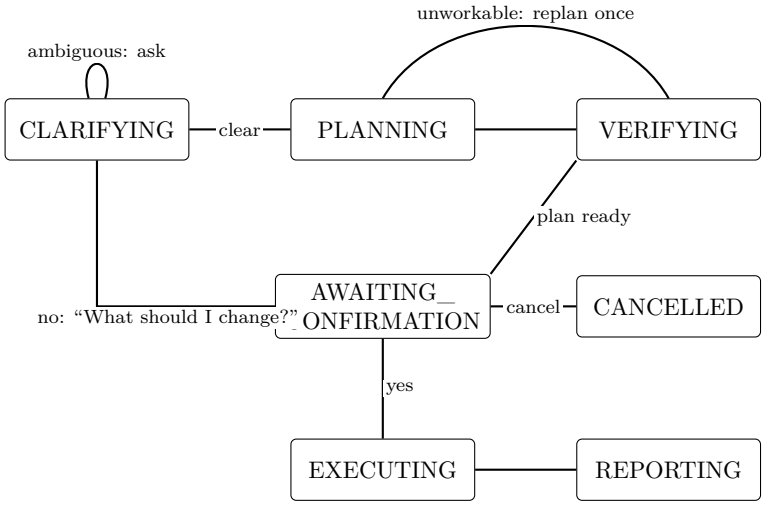


Figure 3.3: The dialogue phases up to the start of a mission. A rejection returns to `CLARIFYING` with the person's reason added to the history.

missed. If the plan checker itself is unavailable, the answer is neither yes nor no, and skipping the plan check is safer than the model hallucinating a concern.

3.4.4 Confirmation

The plan is read back as a short sentence per step. The rover then awaits for confirmation from the user and does not execute any steps until the confirmation is given. The answer provided is classified in two stages. All the keywords are passed over English phase list for all the possible combinations of 'no' and 'yes'. First the reject list is checked before the confirm list,'so 'no, dont't go' is not read as 'yes' because it contains 'go'. If nothing is classified as either 'yes' or 'no'. The rover counts that as unclear and the rover asks for the plan confirmation again.

If the plan is rejected the system asks what it should change and returns to `CLARIFYING` with the answer in the history, so that only one step coul be amended without starting over.

All commands can only be executed vocally however for debug purposes there is a typed input available. It can only be turned on if the environment variables is set for `ALLOW_TEXT_COMMANDS`.

3.4.5 Later Moments

As the rover is moving, the second moment is self-correction. As a step is executed, new frames are compared with the digest. A small module, `revision`, classifies what it sees into the following categories: no change, reroute, a material change or a block. This revision is done without a vision model but normal python code. The planner can be consulted to make a change, but the classifier decided whether the person has to be asked.

When the next step is exectued a reroute is applied. A material change moves the mission into `AWAITING_REVISION_CONFIRMATION`, and the rover waits for a spoken yes before it follows the new plan. The third moment is the report (Section 3.8).

3.5 Failure Recovery

From the pre-video recording which generates the scene digest. New objects can be introduced into the room or old objects can be removed. Situations like an 'orange box' being behind a chair or a bag is left in the rovers path. If the rover reached the desired destination the plan named but the camera sees a different scene. This is a case where the plan was correct but the world did not coporate.

The rover then returns a blocked state, the mission then enters a `AWAITING_GUIDANCE`(Figure 3.4). The rover says why it is stopped using one of these four fixed phrases (target not found, obstacle in its way, arrival contradicted, or it is out of time) or any general sentance and asked what it should do next.

The person can then repond by answering what the rover should do next. The planner can then write a new plan for what it still has to do given the original command, the steps that it already completed, the steps that failed and why it failed. What the rover has already seen and the conversation up until that point. The prompt then tells the planner to use the answer to decide where to look or what to reach instead. If the person says where the target is, the rover should turn that way and approach the target again.

The ouput goes through to checks. The first check is to `validate_step`(Section 3.3). The second approach applies when the failed step was an `approach`. A plan that never approaches anything is rejected.

If both checks pass then the mission carries on from where the rover is, and the rover says 'okay. Trying again'.

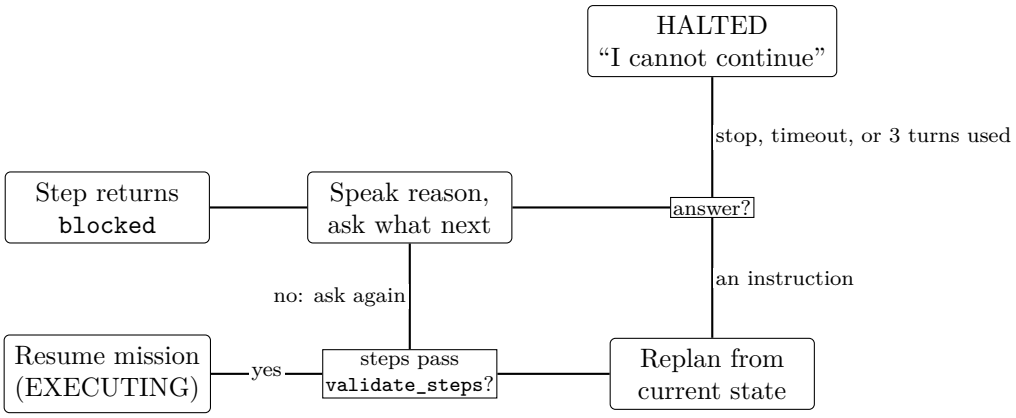


Figure 3.4: The recovery loop. The turn counter is shared across the whole mission, so it cannot be used to drag a failing mission on.

There are three limits that keep this from becoming and open-ended conversation. There are at most three guidance turns per mission. The reply has 120 s to arrive. The person can decline to provide guidance. If the reply begins with. 'stop','cancel','give up' or 'never mind' or. a 'no'. The mission ends with the rover halted.

There are two decisioins that are made here. The first is that recovery fires only for physical or search failures. It never fires for ambiguity. Ambiguity is only asked about before a departure and reported at the end. The reason is to keep the recovery seperated from it is what stops the design from breaking down into KnowNo. The second, is that every each mission is seperated in 1 of 4 categories. completed, completed after guidance, halted after guidance, halted or aborted. This allows us to see what the rover did by iteself or if it was aided by someone. The Report that the rover dispatches also carries a seperate `recovered` flag, and it is a strict flag. The flag is set if an `approach` step succeeded after the last blocked step.

The `halt()` function is not called in a recovery stage. If a step is blocked it stops counting the time it tales to execute that step.The turboPi then sets a `halt()` flag that only the next `execute_step` clears. Since the free-roam approach in Section 3.7.1 directly calls the drive commands on the rover, a flag that is not cleared could cause issues down the line.

3.6 Conditional Commands

A conditional command, which is level L4 test in Table 1.1,asks the rover to make a decision on something and act on the answer: 'check whether the orange box is visible, and if it is, go to it, otherwise stop'. Who actually makes the decision in this case. The planner writes both branches up front, before anything moves, and the rover only evaluates the one condition at the moment it matters. This allows the planner to do the thing it is good at, and turning words into structured plan. Then leaving the single judgements that need a vision model call to the camera and vision model.

Table 3.4 shows the plan shape the prompt asks for. One `check` step states the condition as a short sentence. Steps for the yes case carry `when: yes`, steps for the no case carry `when: no`, and steps with no tag always run.

Table 3.4: A conditional plan, as the prompt shows it to the planner.

Id	Action	Target	When
1	check	the orange box is visible	
2	approach	orange box	yes
3	report	orange box	yes
4	turn	left	no
5	stop		no

When the rover reaches a `check` step in the plan. The rover centers the gimbal, waits half a second for it to settle, and then takes one frame (Figure 3.5). The vision model is asked whether the stated condition is true of what is visible and it can answer yes, no or unsure. Now this loop can remembers the answer. Every later step tagged with that answer is skipped and logged as skipped. The run record now can show which branch ran and which did not.

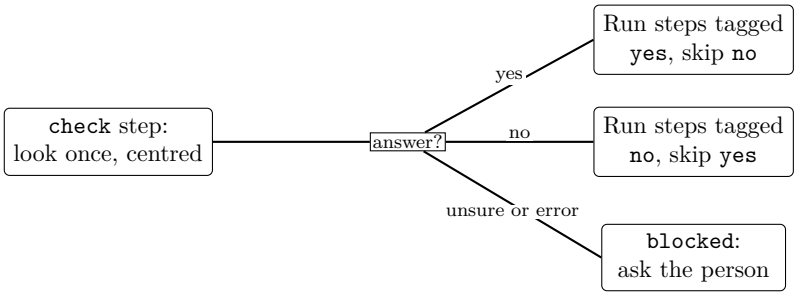


Figure 3.5: How a condition is resolved. An answer the camera cannot support does not pick a branch. It stops the step and goes to the recovery loop.

Now the choice of what happens on the 'unsure', and on a failed vision call. The step returns `blocked`, and the existing recovery loop (Section 3.5) asks the person what to do. There is no default branch added and if the rover guesses 'no' continually it would prompt the rover to go to the incorrect object continueally without anyone understanding if it was working.

There are limits, and I would state them here. The check is one frame from wherever the rover is standing. It does not search. A condition about an object just outside the camera view will come back with a 'no' or 'unsure' even if the object is in the room. The results depends on where the rover is facing.

3.7 Navigation as Instrumentation

Navigation is not what this project is about, but the pipeline needs the rover to go somewhere so that a plan can be a successful or failure . There are two ways of getting there and kept both simple. One follows a taped line. The other finds a named object with the camera and drives to it.

3.7.1 Free-roam Approach

For the rover to successfully arrive at a named target. It needs to approach it. For `approach` the rover has no map and no odometry, so the rover needs to rely on the incoming frames and vision model in order to find the target. The sonar readings and

the final vision model decision decides if the rover has arrived. Figure 3.6 illustates this behaviour

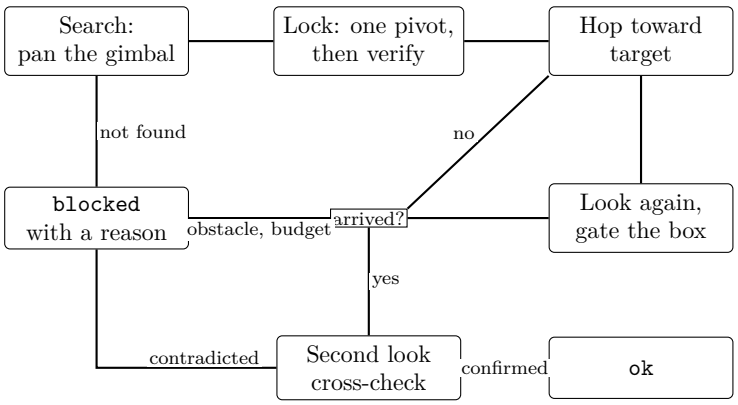


Figure 3.6: The free-roam approach. Each cycle is one hop and one look. Every exit that is not `ok` returns `blocked` with a reason, and that is what starts recovery.

The rover searches the room with the gimbal and camera, not by turning the whole rover chassis. The reason for this is the camera pans through a fixed number of angles for a total of 18 vision model calls in about 3-6 sweeps. The chassis pivots at 110° between each sweep. This is only if the initial sweep does not find anything. Having the gimbal do the scaning is quick since the angle is known from the servo settings. Pivoting the base of the whole rover is a continuous open loop call and it can drift from its path. The vision model reports the target at some horizonrral position in the frame. To get he bearing it i the pan angle plus the offset from the frame center, This is then scaled by the measured 56° field of view. The rover then moves to that exact bearing and the vision model verifies if the object is there.

There are 2 vision models used. The gemma4:e4b model can both do planning and image recognition and qwen2.5vl is a 3-billion parameter vision model that only does image recognition. In each step once an object is tagged it draws a box around it. Every single image captured and processed passes a plausibility gate that rejects a box that fills more than 60% of the frame, boxes with over 95% of the frame's width, a box which is to small to be real, a moving object whose box jumps threefold between hops while the rover is still far away, and a large enough box which sitting exactly at the center of the frame but far away. If the models detects the object as low confidence is held until a second look at the same pan anle puts the box within 0.15 of the frame width than the first look. Any blurry frames with a low sharpness threshold are never sent to the model, since a blurry image will not give any reliable data. The rover will retry to capture the image twice and then skips the look.

Since we do not have any odometry metrics like distance. Each hop that the rover takes is scaled by what the camera and sonar readings are. The default is a fixed pair: A 50 cm hop while the target looks small in the frame, and a 20 cm hop once it fills more than

8% of the frame. A variable mode also asks the vision model for a distance estimate in the same call that it locates the given target. It then hops 60% of the estimated distance minus a 30 cm stop margin. If the target is centered the sonar clearance can shorten that further. Now only a fractions of the estimate is used since the rover hops happen in a open loop and can drift. If the model does not return a usable distance , then the hop falls back to a fixed pair. Both modes were built out since some VLMs estimate distance poorly and some do really well like cloud models. To always have a safety fall back the fixed mode stays the default mode.

The rover decides on arrival by two independent signals. The sonar says it arrived when it reads 300 mm or less on the target center. The camera says it arrived when the box fills 20% of the frame or its bottom edge reaches 90% of the frame height. These are check after atleast one hop or if the bounding box did not shrink and the sonar does not read more than 800 mm. Either of these signals can trigger the vision model to check where the target object is still visinle. If the answer is no, the rover steps back and looks again. It then returns `blocked` with the reasons `arrival_contradicted` if the target stays hidden. I added these checks since a positive feedback from a 3B model, oe from one sonar reading against some other object, is not enough information to call a step complete.

The sonar is a bit complicated, sonar will immediate halt the rover if the rover is less than 300 mm from the object. If the last image taken is the target centered and the rover stops it is treated as the end of the hop. The rover then looks again and checks if the target is still centered and infront of it. A second consecutive stop from the sonar without flagging the rover of its arrival is trated as a obstacle and the step then returns `blocked`

The rover mission has four ways of announcing it ended. First the target is `blocked`, second is the obstacle stopping the rover after trying to do a second hop after coming to a halt, third the arrival image contradicts the cross-check with the expected object and fourth the mission runs our of step cycles which is about (20 cycles or 240 s). Each has a specific reason spoken about in the section 3.5

3.7.2 Scan and Survey

The `scan` command on its own only moves the gimbal, which sees nothing. so therefore us telling the rover to 'tell me what you see' does not have an answer. When surveying the area the gimbal stops the camera at three positions, left, right and centre and takes a frame at each. It has the vision model describe each frame in one sentence. During these scans the rover is stationary the whole time which honours the control splt in Section 3.2.2, since we do not want the vision model to recieve frames which are blurry.

In a Level L5 test command, it asks for more than one sentence. A command like 'what is near the orange box' is asking us to describe an area, and a one-line description loses the small finer details needed. If the command contains any words like *near*, *nearby*,

around, *beside*, *next to* or *everything*, then the rover runs the same three looks after the `observe` or `report` steps. At each of those steps it asks a different question. The prompt goes as follows: List every object in the frame, with its colour and where it sits, and list up to 10 objects. The list is limited due to small vision models usually repeating itself,which can be seen when building the scene digest.

What ever the rover saw throughout its missioin foes into the rport as a list per direction it took. The spoken summary is then asked to name every listed object, grouped by left, centre and right. Each of these perspectives is allowed up to six sentences. The rover cannot look behind it without pivoting 180° so the model bnever lists items it never saw.

3.7.3 The Line Follower

Section 3.2.2 showed that the Mac-hosted models cannot sit in a steering loop. A VLM call takes about 15 s, and keeping a mecanum chassis on a line needs control at 10 to 30 Hz. I therefore left steering to the TurboPi's stock `line_follow` ROS2 node, which closes the loop on the Pi using the chassis-mounted 4-channel IR array. I used the IR array instead of the gimbal camera so the camera stays free for the VLM to look wherever the mission needs, regardless of where the chassis is steering (`rover_pi.py`, `aim()`).

`PiRoverController` does not reimplement steering. It only supervises the node on three things the node has no logic for itself:

- Arrival: the node's own `/line_follow/crossroads_stop` topic decides when the end of the line is reached, not a model's judgement.
- Obstacles: the ultrasonic sensor is polled independently of the line sensor. Two consecutive readings below the stop threshold pause the follower until the path clears, and then it resumes.
- Giving up: a wall-clock timeout. The stock node has no lost-line exit, so without one the rover would drive straight on indefinitely.

I also use a modified variant, `line_follow_corner.py`. When it takes a 90° bend instead of parking at a crossbar, it publishes a `/line_follow/corner` event. This lets a route be given as an ordered string of turns (e.g. `"L"`, `"R"`) and not only the straight-track default. It adds one constraint: a planned pause, such as a mid-drive look, must not land while the follower is mid-pivot. A `STOP` sent during the node's pre-turn nudge cancels the turn instead of the forward motion. `corner_guard_active()` reports a short window after each bend, and callers must defer their pause until it ends.

Halting the follower is where a naive design fails without any error. While running, the node republishes `/cmd_vel` at 100 Hz, so a single zero-velocity `Twist` is overwritten

within 10 ms and the rover keeps driving. `halt()` therefore sends an explicit `STOP` service call to the node *before* the zero `Twist`. It repeats both 0.7 s later, in case the first `STOP` landed inside the node's pre-turn nudge window and was swallowed. This is the most safety-critical part of the hardware integration.

3.8 Reporting

The third ambiguity check is the report at the end of every mission. this happens whether the run is completd, as halted or was aborted. It has two parts. There is a block of facts that is built using the run log. This includes: Which steps ran, ow long each took, how many model calls there were what the approach did, and which models were used. Rules then flag any uncertainty from the same log. This could be a target that was not visible upon arrival. If the vision arrival over a sonar arrival. If the rover try to execute a command outside of its vocabulary. The spoken summary uses a model call which only give the facts and the flags so it had nothing to invent.

The log file is not only important for debugging but also for the Chapter 6, which is why FR9 is a requirement. Every mission ran is written to the disk as a single record that includes the plan, the checks, the guidance turns, light timeline and what model was used for vision and planning. Every condition is checked is listed with its answer. Test levels L0-L5 can also be tagged on mission runs so that they can be grouped without reading every command.

3.9 Summary of Design Decisions

Table 3.5 lists the decisions that shape the rest of the report, with the reason for each and the alternative I turned down.

Table 3.5: The main design decisions.

Decision	Reason	Turned down
Models on a separate machine	About 7 GB of models do not fit on the Pi.	Everything on the Pi, with a much smaller model.
Reactive layer stays on the Pi	A model call takes 15 s, steering needs 30 to 100 ms.	Closing the loop through the vision model.
Room as a text digest	A map is a separate system and out of scope.	Building a map from the walkthrough.
Ask before departure only	The person may not be present once the rover is moving.	Asking when confidence drops mid-mission.
Unclear confirmation means stay	A missed yes is cheap, a missed no is not.	Moving on a best guess.
Vocabulary stated in the prompt and audited	Lets me measure how often the planner leaves it.	Silently correcting every plan.
Vocabulary enforced on recovery plans	Nobody hears a recovery plan before it runs.	Trusting the prompt alone.
Recovery only on failure	Keeps it distinct from clarification.	One mechanism for both.
Unsure condition means ask	A wrong guessed branch looks like a correct one.	Defaulting to yes or no.
Condition action offered only when needed	Keeps the prompt short and stops a condition being invented.	Listing every action in every prompt.
A recovery plan must retry the approach	A plan that walks away from the failed step is not recovery.	Accepting any valid plan.
Gimbal-first search	Pan angles are known, pivots drift.	Spinning the chassis to look.
Two-signal arrival	Neither sonar nor a 3B vision model is reliable alone.	Trusting the first positive.
Fixed hop as default, variable as option	The distance estimate is uncalibrated, and I want it tested first.	Committing to the variable hop.

Chapter 4

Hardware

In the design Chapter 3 was where the conversation went about how to split the decisions between the rover and the brain. In this chapter we will discuss how the rover connects to the brain and how it is controlled by the brain. Translating the plan from the planner into actions a rover can execute. The rover is an off-the-shelf Hiwonder advanced turboPi kit. It uses the open source Robot operating system 2 to execute motion commands.

4.1 The TurboPi Platform

The rover is a Hiwonder TurboPi, Advanced kit, built around a Raspberry Pi 5. The only actuators on this specific unit are a four-wheel mecanum base, which can strafe sideways as well as drive and turn, and a two-servo pan/tilt gimbal carrying the camera. A ultrasonic sensor (sonar) with an RGB ring, sits on the front of the chassis, and a row of infrared line-detection sensors sits underneath the chassis. There is no arm and no gripper on this build.

The Advanced kit does not run a lightweight Python SDK, which is usually on the base robot kit. It can call motor functions directly. Instead it runs ROS2 inside a Docker container on the Pi, and every piece of hardware, driving, the gimbal, the sonar RGB ring, is reached through a ROS2 topic or service rather than a function call. As the rover starts up it lauches `bringup.launch.py`, It launches the `mecanum` node which can turn the velocity command into wheel speeds. Then the important `ros_robot_controller` which. talks to the motor and servo hardware. There is a serial link connected to a STM32 co-processor on the expansion board, a camera connected over USB, a small HTTP video server and The main contact point into the rover is over the `rosbridge_websocket`.

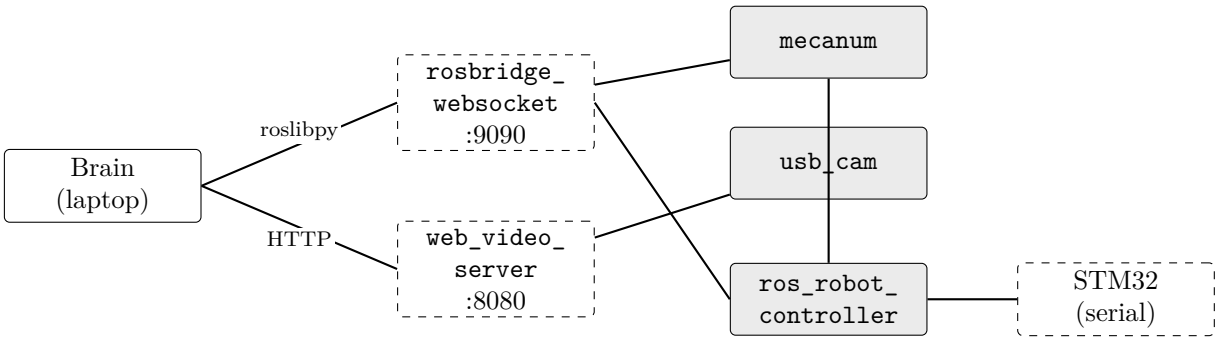


Figure 4.1: What the brain actually reaches on the Pi. Everything above `rosbridge_websocket` and `web_video_server` is Hiwonder's stock ROS2 stack, started as one unit by `bringup.launch.py` inside the Pi's Docker container.

4.2 The Bridge the Brain Actually Uses

The `rosbridge_websocket` is a stock ROS2 package that exposes topics and services as plain JSON over a WebSocket. The brain never needs to read DDS stack or and of Hiwonders Actual packages for control. This was chosen over the brain running `rclpy` since the brain already has existing WebSockets for `/ws/dialogue` and `/ws/execution`

To actually drive the rover you need to send values to the `geometry_msgs/Twist` message on the `cmd_vel` topic. `linear.x` and `linear.y` are forward and strafe speed, and `angular.z` is turn rate. The angular term is summed into each of the wheels kinematics before scaling to a motor speed, so it is not treat as a radian per second value. To figure out what range of values actually turn the rover accurately I chose a range of 3 to 8 which is used in the Hiwonder's own stock code. The gimbal is driven the same way, a `SetPWMServoState` message on `/ros_robot_controller/pwm_servo/set_state` naming a servo id, 1 for tilt and 2 for pan on this unit, and a target position around a centre of 1500.

The frames from the camera are pulled directly from the websocket running on the Pi which had a port of 8080. The endpoint was at `web_video_server`.

4.3 Reusing the Stock Line Follower

Chapter 5

Implementation

In the design chapter (Chapter 3), we went over what each pat of the pipeline does at a higher level. The following chapter says how each of these part is actually built: the libraries, the process, layout, the message formats and how design decisions were turned instrumentation code. There re no specific numbers references in the design chapter. The implementation chapter is where values are defined and implemented.

5.0.1 Speech Input and Output

All the audio files are transcibed using `faster-whisper`, which is a CTranslate2 reimplementation of Whisper, using the `digiphyte/fluister-turbo` model. The model was chosen over a stock Whisper model because it is transcibed to include South African English and Afrikans accents. That is important since I have a South African English accent. The model runs on the CPU with `compute_type="int8"` and 8 threads. The reason for not running it off the GPU is to save on VRAM which can be used for the planner and vision models. Since the audio clips are short enough that the CPU does not become the bottleneck when transcribing the audio.

Audio arriving as an encoded container (webm/mp4/wav) and is decoded to a 16kHz mono PCM waveform via FFmpeg, which is a a multimedia library before being passed to the Whisper encoder. The raw samples are captured directly via sounddevice and can bypass this step. The `transcibe_audio` function takes either a raw bytes (which is an encoded clip form the turboPi, and decoded using FFmpeg) or using a Numpy array of raw float samples. This function is called once per audio clip and not continueally as a audio streams in. This is because the rover does not have to continueally listen to its surroundings for feedback. The audio clip is padded to 30 s internally. Therefore to reduce the number of chunnks you would take on the audio clip.

A empty string is returned in place of all the phrases that are deemed to be hallucinations. It is treated as a failed command rather than letting the planner model fail acting on a hallucination.

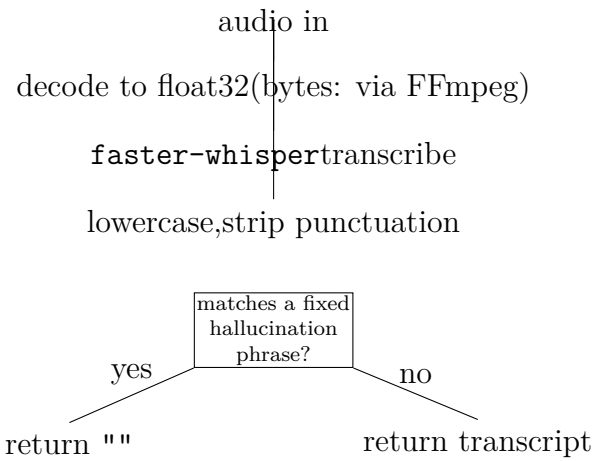


Figure 5.1: `transcribe_audio`. The hallucination check exists because `fluister-turbo` occasionally transcribes silence or background noise as one of a handful of fixed captioning artefacts, most often "thank you so much for watching".

5.0.2 Speech Output

For someone to actually hear the rover speak back. This goes through Piper, a local neural text to speech engine. which uses a `een_US-lessac-medium` voice. The `tts.py` mirrors stt.py's way of caching-model pattern: `get_voice` will load the PiperVoice model once and keep it as a callable instance. Piper's voices are not fetched automatically the same way `faster-whipers`'s model is. Initially I had to download the voice using: `python -m piper.download_voices en_US-lessac-medium`. Then point the environment variable at the `.onnx` file as well as its `.onnx.json` config file. There is no Afrikaanse voice available so the models were only configured in english.

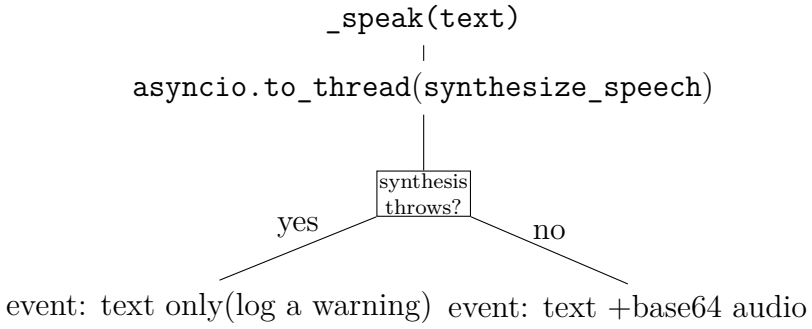


Figure 5.2: Synthesis runs on a worker thread because `_speak` shares an event loop with `ingest_frame`: synthesis is sub-second but still blocking, and perception should not stall mid-mission for a voice line. A failure degrades the mission to text-only rather than stopping it, the same soft-fail stance the pre-departure dialogue's own `_speak` in `pipeline.py` takes.

5.1 Model Inference Layer

5.1.1 Planner

There is a `InferenceProvider` file which is an abstract class with one method, complete. This method takes lists of chat messages which as disguised as prompts as well as a optional image parameter and returns a `Completion` which has the following output. The text, which model answered, the latency as well as tokens used as well as the cost which is seen in the report. There are currently wo providers local and cloud models. The local provider is called `OllamaProvider` and the cloud provider called `OpenAICompatibleProvider`. Both providers implemented the same Input and output method described above. It is used since it is easy to interchange between cloud and local models.

Every provider creation is cached by (role, provider, model), so a planner can be reused instead of reconstructin it on every call. Swapping the planner or vision model is a matter of an environment variable (`PLANNER_PROVIDER`, `VLM_PROVIDER`), not a code change. This stays in-tune with my modular design so testing different local models comes with ease.

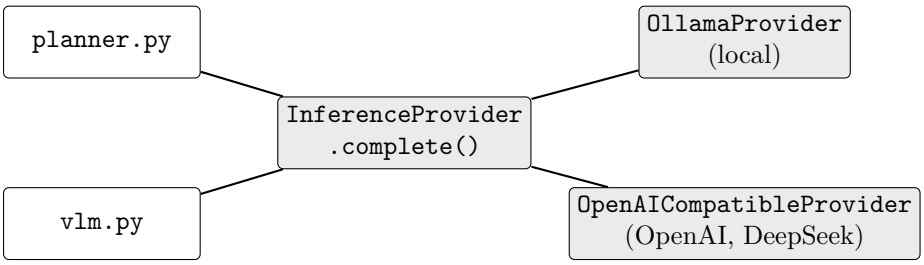


Figure 5.3: Both roles call the same interface. `get_provider(role)` resolves `PLANNER_PROVIDER` or `VLM_PROVIDER` to a concrete class and caches the instance.

The `OllamaProvideer` sends `num_ctx` I defaulted this value to 8192 which I explain why in the Chapter 6 An optional `repeat_penalty` parameter is allowed, and `format="json"` for every planner and VLM call. Ollama allows you to call a model twice before it fails the missions. Two crashes which I ran into was a cold CUDA load on `gemma4:e4b` specifically and a small model loops on the same tokens. `build_plan_prompt` function assembles a prompt form the active planner profile (Section 3.3), the room digest when one is loaded, and the conditioanl guidance block when the command contains a conditional word (Section **??**). A pure function which is seperate from `generate_plan`. The following `tools/snapshot_ptompt.py`, can print out the exact prompt a command would produce without calling a model.

Having a look inside the Failure recovery (Section **??**ssec:guidance*loop*)).*Thereisansecondpromptwhichisbuiltusingwhatwecreatedcalled*,`build_r`

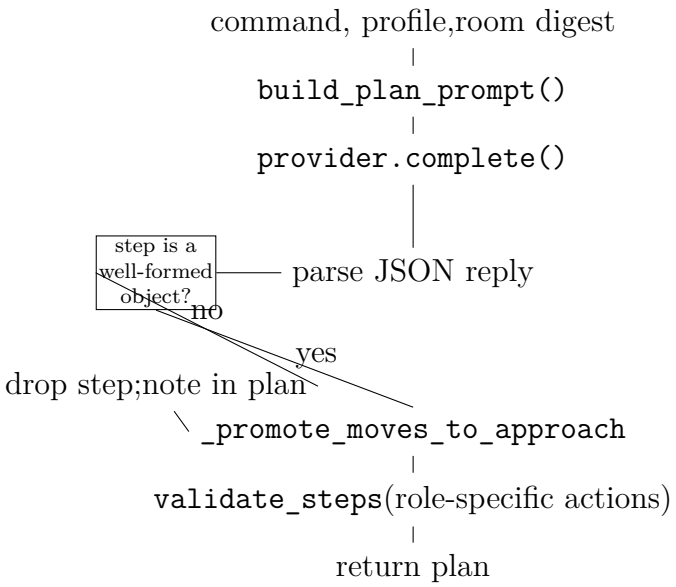


Figure 5.4: `generate_plan`. A smaller model sometimes writes a step as a bare string instead of an object, or nests the list oddly, so every step is checked before anything downstream calls `.get` on it. `_promote_moves_to_approach` rewrites a `move` step whose target names an object into an `approach` step (Section 3.3); the retried transient failures from `provider.complete()` are omitted here for space and described in the text above.

5.1.2 Vision-language Verification

All the calls which are made to a the local vision model go through one function, `_ask`. This function takes a named stage, a prompt, an optional image and a salvage flag. It then calls the correct VLM depending on the provider, cloud or local. Ollama also comes with a parameter to set the output format to json using, `json_mode=True`. Each previous function is a single shot prompt which then calls the `_ask` function to execute that step. The single purpose prompts are all functions namely: `check_ambiguity`, `verify_plan`, `describe_frame`, `inventory_frame`, `locate_target`, `check_condition` and `survey_frame`. None of the prompts used follows a particular template as this can results in the answer from the model having extra fields which are not intended.

The `locate_target` has the most moving parts. It is called once per cycle in the free-roam approach (Section 3.7.1). The prompt asks the VLM for a bounding box, how confident it is in it's answer and how far away the object is. Using `distance_cm_from` it converts whichever unit the model used inot centimetres, or if it fails it returns `None`. Any distance estimate that falls outside of (0, `VLM_DIST_MAX_CM`], including a `NaN`, is also treated as if it were `None`. Any value that the model outputs is handled the same way. Since a model this small is small the predictions can be slightly incorrect

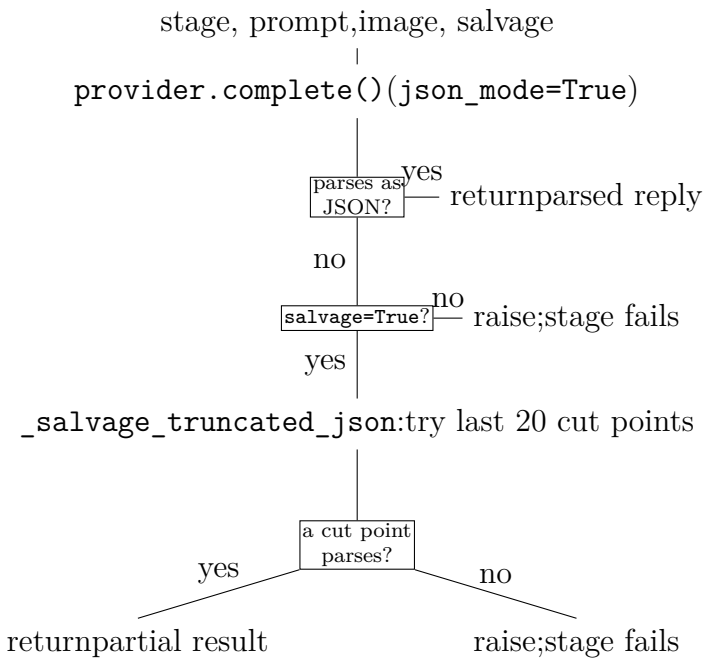


Figure 5.5: `_ask`. Salvage is only enabled for `inventory_frame` and `survey_frame`, the two calls that ask for a list of objects and sometimes hit `num_predict` mid-list on a model that has started repeating itself; a reply with ten correct objects and one truncated one is more useful than discarding it. Every other call keeps `salvage=False`, since a truncated yes/no answer or bounding box is not something a partial parse can safely recover.

5.2 Clarification and Mission Orchestration

5.2.1 Clarification Manager

The clarification loop (Section 3.4) is a small state machine inside of ,`DialogueSession`, with `Phase` as enumerator: (`CLARIFYING`, `PLANNING`, `VERIFYING`, `AWAITING_CONFIRMATION`, `EXECUTING`, `REPORTING`, `CANCELLED`). There are also functions inside of `pipeline.py` which change the current Phase in which the rover is in. This is built this way so that everyfunction inside of pipeline reads the phase before deciding what to do with incoming messages from the rover. If boolean flags were used this would result in a state being missed or miss interepeted especially since most function calls inside the mission files make use of worker threads.This would result in flags being set errouneously.

`effective_command` helps to remove all the noise in the raw logs. Now the reason for there is three checks for the command at different stages. The resolved command once the vision model has written one, the orginal command spoken if the dialogue sessions reaches its cap and the current command with the dialgue after all the clarification questions. Using only 1 function to update the command resulted in the dashboard, the planner and the logs always having the same updated command.

The confirmation file has a smaller state machine inside of `confirmation.py`, and this

is seperated from the normal Dialogue session. Since its only job is to turn whatever was said into a yes, no or unclear output. The `keyword_pass` function checks the reply given again English and an Afrikaans phrase list. The rejected list is check first so a phrase like 'no, don't go'. Is not classified as 'yes' to go. This part of the pipeline does not need a model to interpret what it should do. Since either the rover should go, 'yes', or It should stay, 'no'.

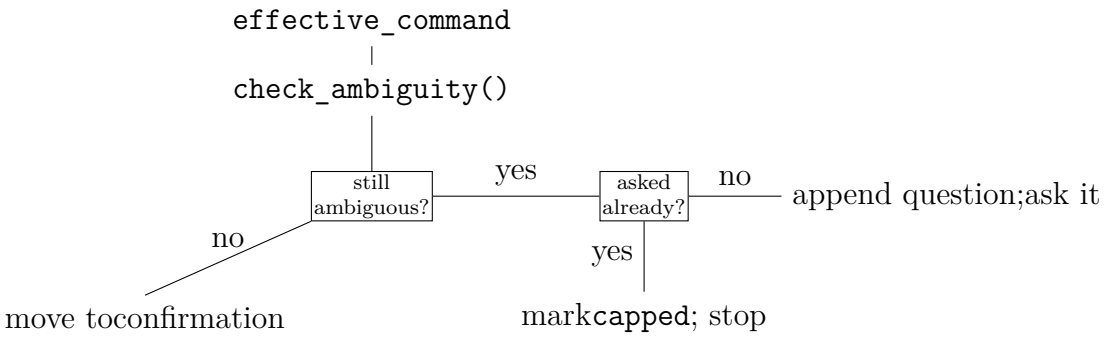


Figure 5.6: `_advance_to_confirmation`, one turn of the clarification loop. The repeat-question check exists because a model that cannot think of anything new to ask would otherwise repeat the same question until the five-turn cap was reached, rather than giving up as soon as it had nothing left to add.

Once the back and forth dialogue between the rover and the human is done. The `DialogueSession` hands the mission state off to `MissionSession`. Once that is complete the rover is no in a new state of executing and will only communicate back to the user if it needs help with guidance or if it has arrived. The state machine at play here is mission phases.

5.2.2 The Mission Execution Loop

Once the plan is confirmed, `run_mission` is one function that drives the rover, one step at a time, until the mission reaches its terminal phase or it ran out of approach steps. This is kept as one function for simplicty since if it was written in a class with multiple methods it would complicate it more. It does 4 things in a loop. Read the cursor, run one step, decide what the results mean and move to the next step. If it was written in a class it would mean passing in the arguements between different methods and that would be more debugging work.

Which function actually runs a step depends on what the step is and on what the rover object in use can actually do, and that second condition matters just as much as the first. A `check` step goes to `_run_check`. An `approach` step goes to `approach_controller` (the free-roam hop loop, Section 3.7.1), but only when `NAV_MODE` is `free` and the rover object exposes `pivot`, `hop`, `get_frame` and `set_gimbal`. Everything else, including every `approach` step when free roam is off, goes straight to `rover.execute_step` on a worker

thread. To validate if a specific rover that is connected has a specific function. We used the `hasattr` function, since the `VirtualRover` and the `PiRoverController` has different availablility. E.g The virtual rover does not have a camera that it can pan and trying to code that would not solve our initial problem.

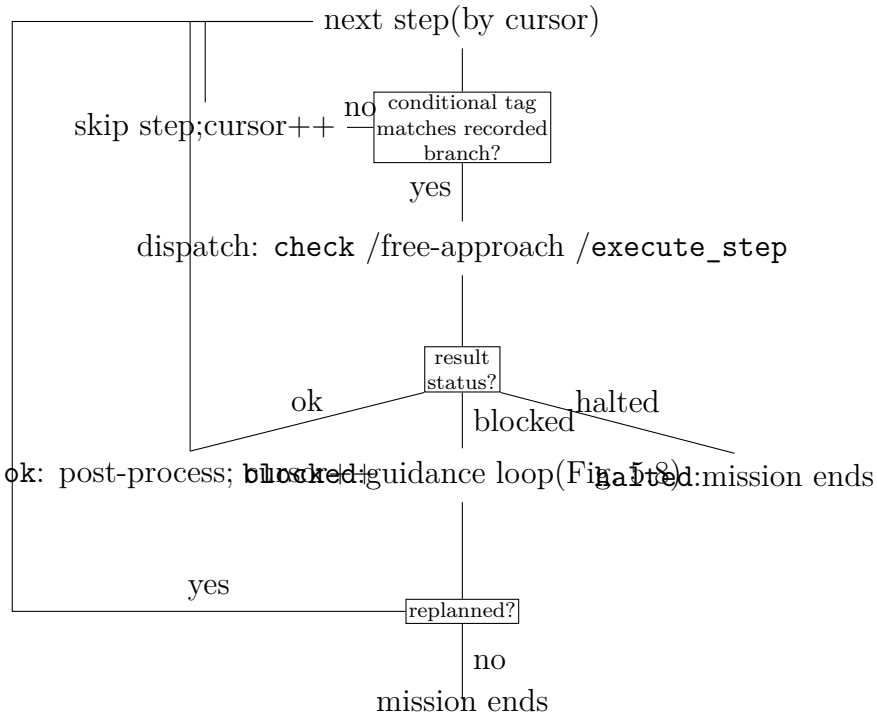


Figure 5.7: `run_mission`'s step loop. The conditional-tag check is what Section 3.6 relies on: the planner only decides which steps carry a `when` tag, and this check is what honours it. The loop's own `finally` block (not shown) writes the report for every non-aborted outcome; an operator-triggered `asyncio.CancelledError` skips straight to `rover.halt()` instead.

If a result is blocked it is handed off to the guidance loop (Section 5.2.3). A `halted` results ends the mission immediately with no guidance offered, this comes down to the step itself deciding that the mission cannot continue. This could be something in the way of the rover. If the mission was not aborted it will write its report in the loop's `finally` block. It does not matter if it finished cleanly or ran out of guidance, or it was halted outright. All runs are tagged to be able to compare the results of why it was halted. The only path that skips writing an report is `asyncio.CancelledError`. It calls the `rover.halt()` immediately and writes the determinitsitc half of the report straight to the disk with `spoken=False`, stopping the rover is never delayed by a planner call or spoken summary.

5.2.3 Failure Recovery in Code

If the rover is `blocked` it is handed to `_guidance_loop`, which is the recovery behaviour of Section 3.5 is implemented. The rover stops immediately if the `GUIDANCE_ENABLED`

flag is off. This is for testing purposes only. The guidance loop then builds a question from a fixed template keyed on the failure reason (`target_not_found`, `obstacle`, `arrival_contradicted`, `cycle_budget_exhausted`, or a generic fallback) and it asks the user up to 3 times on what its next steps are. The questions for guidance are templated and not generated by a model for the simple fact that the model might have planned incorrectly so it is not a good idea to ask the source of the error why It has stopped. This makes it predictable to know that the rover is lost and needs the actual steps to get to its target.

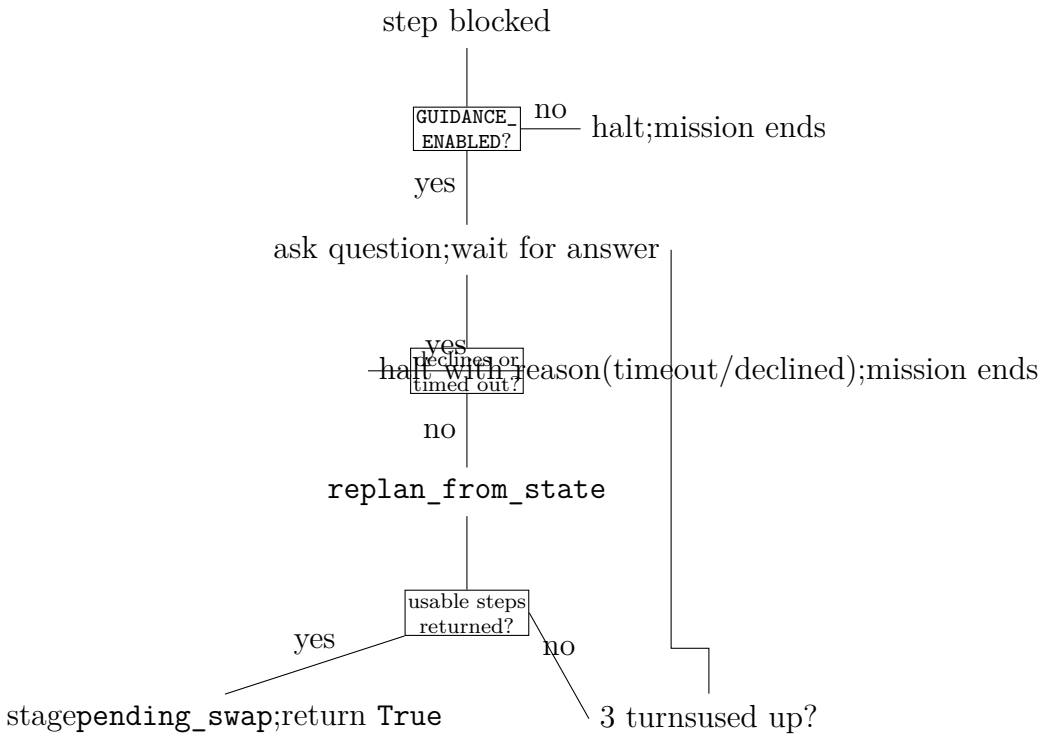


Figure 5.8: `_guidance_loop`. A "no" out of the turn-count check on the right loops back to ask again rather than ending the mission; running out of turns is itself a third halt reason (`guidance_exhausted`), alongside timeout and decline, so a batch of failed runs can be told apart by why they failed.

5.2.4 Mid-mission Revision

The frames sent from the rover during a mission is sent through `ingest_frame`, which has 3 checks to limit the amount of calls to a bigger model. The VLM is not asked to look at every single frame. this is used since the rate at which the camera produces frames is higher than the rate at which the VLM can process images. The three tiers in Figure 5.9 let cheap, local checks filter out the frames which are very similar and the VLM can prioritize the frames which are unique.

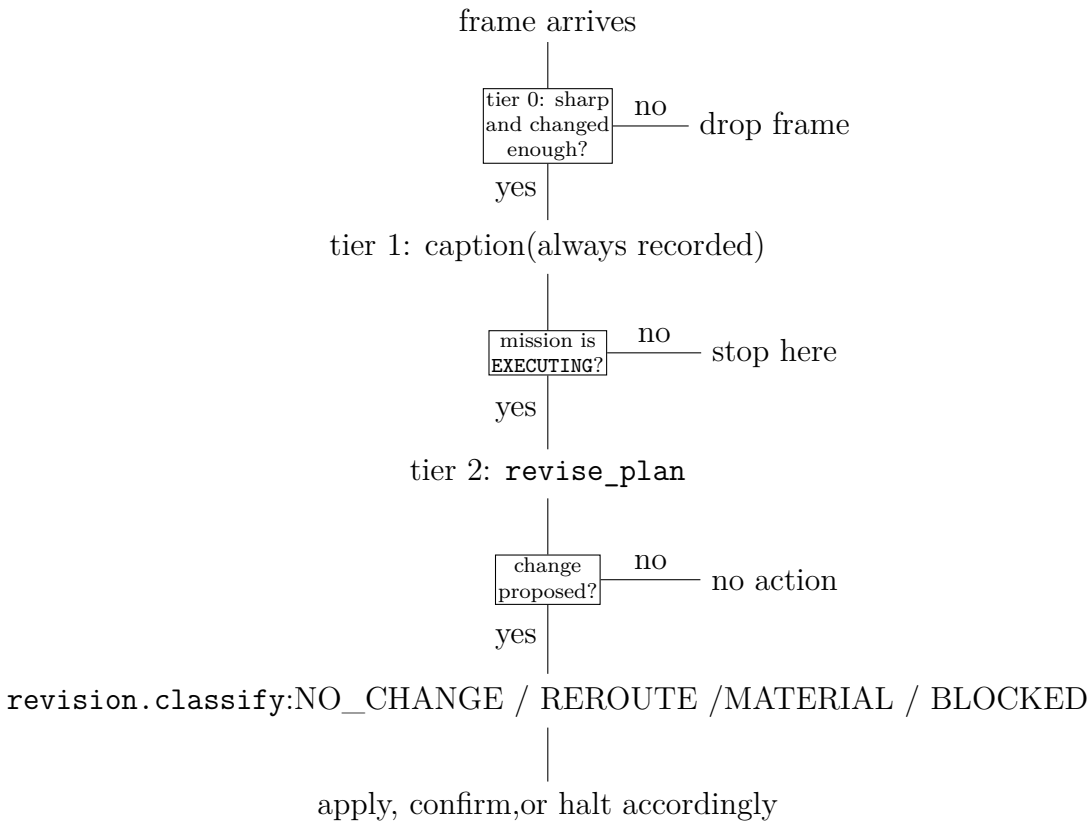


Figure 5.9: `ingest_frame`'s three tiers, feeding `revision.classify`. Tier 0 is a 32×32 greyscale thumbnail diff plus a Laplacian-variance blur score (`PERCEPTION_MIN_SHARPNESS`, `PERCEPTION_MIN_INTERVAL_S`, `PERCEPTION_CHANGE_THRESHOLD`); tier 1 is one `describe_frame` caption; tier 2 only fires while executing.

The `revisioin.classify` is a plain rule, no model calls at all. It compares the confirmed plan's targets against the proposed plan's. Using the action vocabulary for the specific rover. If you add a target, removing a confirmed target or a new action is introduced that is outside of the fixed navigation whitelist of tartgets. These all count as `MATERIAL` changes and stops the mission for a yes confirmation. Then `handle_revision_confirmation` applies it. If there is an empty revised plan is `BLOCKED` it ends the mission as well since there is nothing to revise. If the goal and every target is unchanged, and the only new actions are navigation such as a different route around an obstacle. This is treated as a small `REROUTE` that the rover can follow without confirmation.

5.3 Rover Client and Reactive Layer

5.3.1 Real Hardware

Inside of the `RoverController` is an abstract class with two required methods, `execute_step` and `halt`, as well as a optional `status_light`, which is only available

on rovers which have the status light hardware installed. This functionality is simple. Every rover implementation has to run a step and has to be able to stop. There are interfaces the physics-free simulator, and a rover which communicates over ROS2 over a network socket. There are two main rovers the `VirtualRover`, describe in Section 5.3.1.1, and `PiRoverController`, which controls the real TurboPi using roslibpy.

The `PiRoverController` opens one `roslibpy.Ros` connection, advertises a `/cmd_vel` topic for driving and a PWM-servo topic for the gimbal. It also opents two ROS2 services to access `/line_follow/cmd` and `/line_follow/set_runnning`. These services control the stock line-follower node rather than reimplementing line following from scratch.

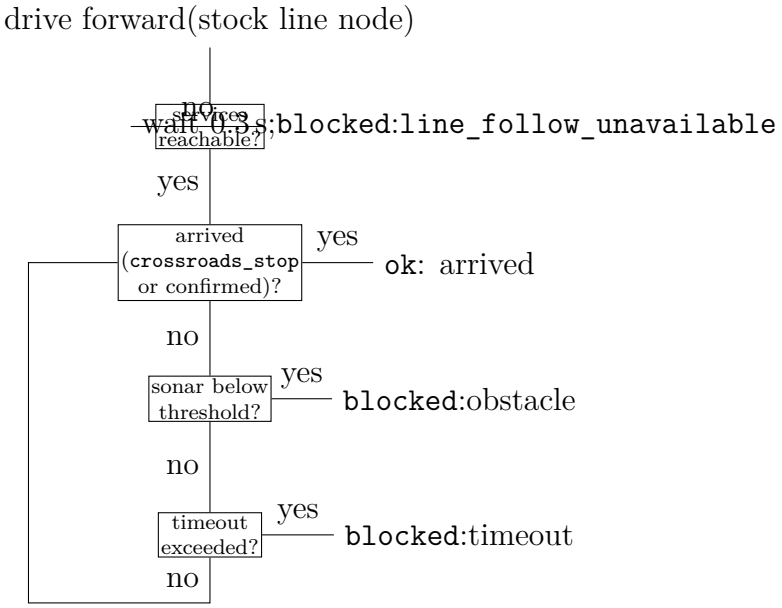


Figure 5.10: `_follow_line`, the supervisor added around the stock line-follower node. It owns exactly the three things the node has no opinion about: arrival, obstacle stopping through the polled sonar (using `_sonar_seq` to tell two fresh readings from one stale one read twice), and a wall-clock give-up, since the node itself drives forward indefinitely if the line is lost. The `_lf_unavailable` check (added 11 October) means a drive that never started is reported as blocked rather than as a drive that ran and found nothing.

5.3.1.1 Virtual Testing

This is where almost every piece of evaluation data comes from. the `VirtualRover` is what every single run without physical hardware actually executes on. The virtual rover is built around one idea: `approach` is a navigation skill, It plans a routes and follows it, while the normal `move` and `turn` are commands which execute as give like drive forward, or turn right. The approach step hands over the target to the rover and expects it to work out how to get there, while a `move` and `turn` step only hands over a specific distance or angle.

Both halves read from the same OccupancyGrid (nav.py), built once per room with the rover's footprint clearance baked in, so the planner and the collision checker can never silently disagree about what's free.

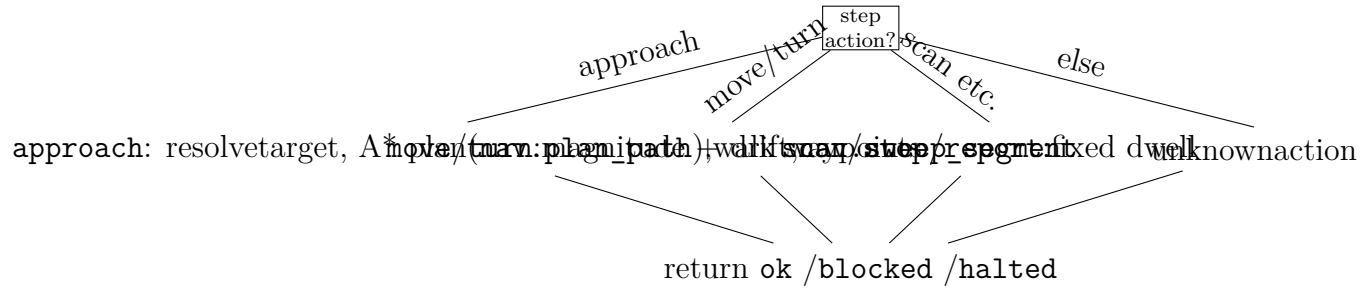


Figure 5.11: `VirtualRover.execute_step`'s dispatch. An `approach` step can only fail by planning (no path, or an unresolved target logged as a hallucination for RQ2), never by collision; a `move` or `turn` step is swept against the same grid and stops at the last free point if blocked, so a collision there means the chain of plan steps itself went somewhere it should not have.

The room itself, a `RoomMap` loaded from JSON (`room_map.py`), is hand-typed, not estimated. I measured coordinates for every named landmark off a sketch and a tape measure once per test room. A landmark's point is the standing spot a rover can occupy to look at the thing. This is needed so the navigation code knows where the rover should end up. The simulated and wall-clock time are kept apart the same way: every action's duration is computed in simulated time first and only then divided by `VIRTUAL_TIME_SCALE`. So a sweep of many runs can be played back faster than real time without changing the distances, speeds, or timings the evaluation actually reports on.

Every run writes one JSON line per step to a file to trace it and `summary()` is able to derive statistics used for the second research question. Things like how many steps counted, how many steps were blocked, how many targets were hallucinated and which ones. These results are logged after each step rather than after the final outcome. Since a single pass/fail number does not contain the why it has that certain outcome.

5.4 Communication Protocol and Dashboard

The brain is a FastAPI application backend server. The two endpoints that matter for a mission, `/ws/dialogue` and `/ws/execution`, are WebSockets carrying small JSON frames rather than a request-response API, since both sides need to push events the other did not ask for, the brain needs to tell the Pi client a step started without the client polling for it, and the client needs to tell the brain a frame arrived without the brain asking for one on a timer. A request-response API would have meant one side constantly polling the other for events that have no fixed schedule, which is wasteful and adds latency. This is a

message protocol rather than a branching algorithm, so Figure 5.12 below is a sequence diagram, not a flowchart: it shows who sends what, not a decision to be made.

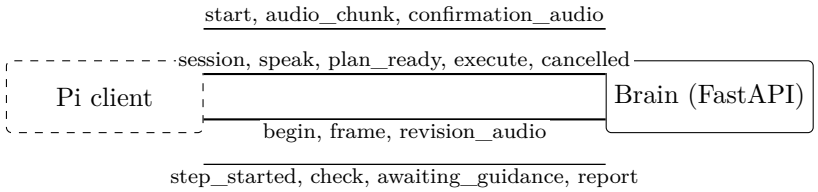


Figure 5.12: `/ws/dialogue` (top pair) carries everything up to confirmation; `/ws/execution` (bottom pair) carries the mission. One frame per complete utterance, not a stream.

Both connections are opened with `max_size=None`. This was not the original setting. A long spoken plan read-back, a `speak` event carrying the base64 WAV Section 5.0.2 describes, exceeded the websockets library's 1 MiB default size and closed the connection. A few smaller endpoints round out the API. The `/scene` endpoint accepts an uploaded walkthrough video and returns a scene id. The `/logs` endpoint and `/logs/{session_id}` endpoint serve the run log listing and detail that the dashboard's Logs page reads from. There is a small `/logs/{session_id}/flag` endpoint that lets an operator mark a run good, bad or for review with a note

The dashboard is a React and TypeScript single-page app, a thin client over these same endpoints, and it holds no logic of its own that the evaluation depends on, every number Chapter 6 reports comes from the JSON logs, never from anything the UI computed on its own. It is a debugging and demonstration tool and nothing more. Something that lets me upload a room walkthrough video, watch a dialogue and a mission happen panel by panel, and drive the rover manually, all without needing a terminal open

5.5 Logging and Measurement Instrumentation

There are two logging methods used here. The `brain.log` is a conventional application log, written through Python's `logging` module,interleaving every call in time order with full prompts and raw model output at debug level. Then we have the run log, one JSON file per executed mission. This is the run log our evaluaiton secition gets the information it needs. The `report.py` file assembles: he plan, every check and its result, the guidance log, the light-state timeline, the model tag for both roles, and a tagged `test_level` when one is set.

Every model call also writes one entry to a smaller usage ledger (`providers/usage.py`), holding latency, token counts and cost when the provider reports them, and this is reset at the start of every session. A separate, offline tool, `tools/context_budget.py`, measures something the run log on its own cannot. How many tokens a given prompt would cost

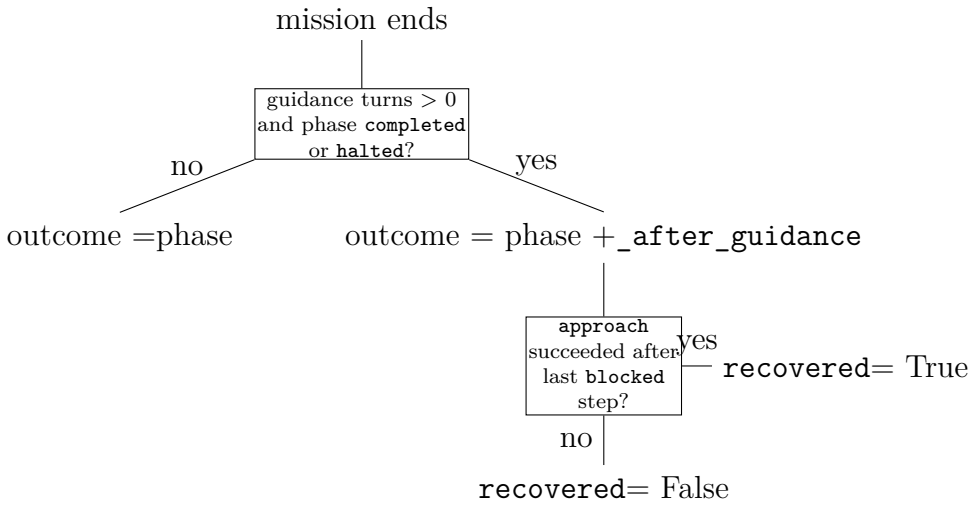


Figure 5.13: `report.py`'s outcome classification. A mission that needed at least one guidance turn is never reported as a plain `completed`, even if it finished, because the distinction between a clean run and a recovered one (Section 3.5) is exactly what RQ1 needs to see; `recovered` only counts a run that actually reached its target afterwards, not merely one that ended without halting.

before it is ever actually sent to a model. If you ask ollama for the `prompt_eval_count` and send in a prompt with the parameter `num_predict=1`. It counts the tokens the model would have to read without paying for a real generation afterward. It can measure the fixed instructions, t he room digest, one clarification turn and one recovery turn separately. This allows you to see a prompt's token budget is actually going before running a mission to see if it would fail.

Chapter 6

Experimental Evaluation

This chapter reports what the pipeline did in two kinds of test. The virtual tests drive a simulated rover through a hand-drawn room map using typed commands, and they measure how long a chain of instructions can get before the plan stops matching the command (RQ2). The physical tests run the voice pipeline on the TurboPi in a real room, and they show which command levels in Table 1.1 the pipeline can carry out (RQ1). Every number below comes from the run logs produced by the pipeline itself, and the sample sizes are small enough that I report counts, not percentages.

6.1 Experimental Methodology

6.1.1 Command Corpus Design

Virtual tests. The virtual rover moves through a hand-authored map of one room (`room_tour1`), with every position expressed in a fixed metric frame and drift modelling switched off, so that any failure belongs to the plan and not to the simulated odometry. Commands are typed, and the audio loop is tested only on the physical rover. Four kinds of chain were generated, and each is scored against the sequence of landmarks the command asks for:

- *approach* chains of named landmarks ("go to the couch, then the coffee table, ..."), at depths of 1 to 12 instructions;
- *literal* chains of metric moves and turns ("move forward 0.6 metres, turn right 90 degrees, ..."), at depths of 2 to 8;
- *mixed* chains that combine the two, at depths of 4 to 8;
- *probe* chains of depth 8 that contain one step the room or the rover cannot satisfy: a landmark that is not in the room, or an action outside the capability vocabulary ("pick up the lamp").

Each command was run five times on the local pipeline and three times on the cloud benchmark. A separate context sweep (Section 6.4) used approach chains from depth 4 to 40 in steps of 4, three runs each, at three context settings.

Physical tests. The physical tests use the command levels of Table 1.1, with a level 0 control added in which the rover is only asked to describe what it sees. Each level has a small fixed set of commands built around two objects on the floor of the test room, an orange box and a black bag (later also a blue bottle). Commands are spoken, transcribed by the speech-to-text model, and confirmed by voice before the rover moves. The rover is never given a map or any marker, so every target has to be found by the camera.

6.1.2 Evaluation Metrics

For the virtual tests, a run counts as *correct* when the plan visits exactly the expected landmarks in the expected order (`visits_match`). A run is *completed* when the mission reached the end of the plan, and *halted* when the rover stopped on a blocked step. A *hallucination* is an action, object or location in the plan that is not supported by the command, the room or the rover's capabilities, which is the definition used in Chapter 1. Collisions with virtual obstacles are counted separately.

For the physical tests, each run ends in one of four outcome classes recorded by the pipeline: `completed`, `completed_after_guidance` (the rover stopped, asked what to do next, and finished after the user answered), `halted_after_guidance`, and `aborted`. A run counts as handled when the logged plan matches the command and the rover reached its goal without intervention. I did not score the physical result of every run beyond the log, and Section 6.2 marks where that matters.

6.2 Plan Correctness under Vague Commands

Table 6.1 summarises the final physical batch: 23 runs on the TurboPi with Gemma 4 E4B as both planner and vision model, recorded on 10 and 11 October.

Candidate: L5 has only 2 runs; update this row and the total after the final two L5 runs, and add the raised-target (lamp on the desk) case.

Levels 1 and 2. All six runs produced the expected plan on the first attempt (`turn left, move forward` and `move forward, turn right, move forward`) and finished in 2 to 3 s. These levels need no perception, so with three runs per level they show that the planner maps simple action chains onto the capability vocabulary without help, but they do not give a failure rate.

Level 3. The plan shape was right in every run, because the pipeline turns "drive forward until you see the black bag" into an `approach` step followed by an `observe` and a `report`.

Table 6.1: Physical rover outcomes per command level, final batch (Gemma 4 E4B for planning and vision).

Level	Type	Runs	Done	After guid.	Halted/ aborted
L0	Describe scene (control)	4	4	0	0
L1	Single action	3	3	0	0
L2	Sequence of actions	3	3	0	0
L3	Perception-dependent	4	1	2	1
L4	Conditional	7	6	0	1
L5	Vague	2	2	0	0
	Total	23	19	2	2

Execution was less clean. Two of the four runs needed guidance (one needed three turns), and one halted after guidance. The halted run was the command "please don't do the black bag", which the pipeline resolved to "drive to the black bag" and then drove, so the negation was lost. The same batch contains transcripts that were visibly corrupted by the speech-to-text stage ("arif box" for the orange box, "go towards the orange box if you can see it and right to hear it"), and two of the four level 3 commands were of that kind.

Candidate: Check the spoken wording for these four runs; if "don't" was actually said, report it as a negation failure, otherwise as a transcription error.

Level 4. Five of the seven plans encoded the condition as an explicit `check` step followed by the branch (for example `check, approach, observe, report, stop`), and in those runs the rover either proceeded (three runs) or stayed put (two runs). Two plans dropped the condition entirely: the command was rewritten to "go towards the orange box" and the rover simply went. One of these two, with the black bag as the target, was aborted at an obstacle. Counting a dropped condition as a failure, five of seven level 4 plans were faithful to the command.

Candidate: Add whether each branch decision matched the true scene (box next to bottle or not) for each run; I cannot score this from the logs. Also drop or flag the run where the operator said where the box was.

Level 5. Both final-batch runs completed, but neither is a strong test of vagueness. One names the target outright ("tell me what is near the orange box"), and the other was a garbled command ("go towards the rn fox and then come to see a rally") that the clarification loop turned into "go to the orange box and tell me what you see around it". The clarification step is what made this run work, which is the behaviour the design intends, but a single case cannot support a rate.

Exploratory runs. Earlier runs between 7 and 9 October used different model pairs and are not part of Table 6.1. In six search-and-approach runs with Gemma 4 E4B as the vision model, five halted after guidance with the target not found and one completed. In seven search-and-approach runs with Qwen2.5-VL 3B as the vision model, three completed directly, two completed after guidance, and two aborted. Six further runs on 9 October

(levels 0 to 4, Qwen as the vision model) gave four direct completions, one completion after guidance and one halt after guidance. One of those completions is excluded because the spoken "Thank you." was resolved to the previous command's plan, which shows that the dialogue state carried over between runs. These runs differ in command, planner and scene, so they are not a controlled comparison, and the final batch reached its targets with Gemma as the vision model, so the 8 October failures were not a permanent limit of that model.

Candidate: State what changed between 8 and 10 October (prompt, thresholds, scene), since the vision model alone does not explain the difference. Also say whether the dialogue state carry-over is fixed.

6.3 Executability on the Physical Chassis

Because the rover has no wheel encoders and drives in short stop-look-hop cycles, two measurements decide whether a plan can be carried out safely: how well the ultrasonic sensor stops the rover, and whether the vision model can supply a distance estimate for the hop length.

Ultrasonic stop. The stop threshold was set to 300 mm. Table 6.2 gives the sonar reading after the rover stopped. When the rover approached a box square-on or a wall, the reading after the stop was 184 to 291 mm in all twelve runs, which is up to about 12 cm inside the threshold because the sensor is polled at 100 ms and two consecutive readings are required. The sonar did not see a box approached at 30° to the axis: it read 875 and 1097 mm after the stop, so the beam passed beside the box. A thin chair leg stopped at 426 mm. A low obstacle approached at an angle is therefore the case the ultrasonic layer does not cover, and the vision check is the only protection there.

Table 6.2: Sonar reading after the rover stopped, per case (mm). Threshold 300 mm.

Case	Runs	Sonar after stop (mm)
Box, square-on, from 50 cm	3	203, 231, 211
Box, square-on, from 30 cm	3	239, 291, 237
Box, square-on, from 20 cm	3	184, 203, 200
Wall	3	206, 218, 206
Chair leg	1	426
Box at 30° off-axis	2	1097, 875

Candidate: The sonar bench CSV stores the tape-measured gap in a column named gap_cm, but the values (180 to 291) match the sonar readings in millimetres, so the recorded pass flags (15 of 15 against a 25 cm limit) cannot be trusted. Confirm the units you typed, and recompute the pass count before citing it.

Distance from vision. Section 6.2 relies on the vision model locating the target. I also tested whether it can report how far away the target is, using five true distances from 30 to 150 cm with three frames each. The target was not visible in any frame at 30 cm for the Gemma model and in all three for Qwen in numeric mode, because the camera is too low to see an object that close. For the other distances, Table 6.3 shows that neither

model gave a usable absolute distance. Qwen2.5-VL 3B returned only two values (0.5 m and 1.5 m) in numeric mode, and in bucket mode it returned the same "more than 2 m" bucket for every visible frame. Gemma 4 E4B tracked the ordering of the distances in numeric mode but overestimated by 1.25 m on average, and in bucket mode its ranking was uncorrelated with the true distance. The variable hop length based on these estimates was therefore not adopted, and every mission in the final batch used the fixed hop.

Table 6.3: Vision-model distance estimates against measured distance (visible frames only). MAE is the mean absolute error; the rank correlation is not defined for Qwen in bucket mode because its output was constant.

Model	Mode	Frames	MAE (cm)	Rank corr.
Gemma 4 E4B	numeric	12	125	0.77
Gemma 4 E4B	bucket	11	49	−0.12
Qwen2.5-VL 3B	numeric	12	40	0.43
Qwen2.5-VL 3B	bucket	13	161	n/a

6.4 Compositional Depth and Hallucination Onset

Table 6.4 gives the virtual results for the local pipeline and the cloud benchmark. Across 105 local and 63 cloud chains of the approach, literal and mixed kinds, every plan visited the expected landmarks in the expected order. No hallucination was recorded in any of these 168 runs, so within the depths tested (up to 12 instructions for named landmarks and 8 for literal and mixed chains) I observed no onset of hallucination.

Where execution failed. The only non-completed non-probe runs are the depth-8 mixed chains, where the plan was correct but the mission halted. The command "... go to the bookshelf, turn left 90 degrees, move forward 0.6 metres, turn around, move forward 0.6 metres, ..." produced a plan that followed the wording exactly, and the literal 0.6 m move after the bookshelf collided with a virtual obstacle in five of five local runs and one of three cloud runs. The planner noted this risk itself ("the 0.6 m straight movement ... may encounter an obstacle because `move` has no obstacle avoidance"). This is a plan that is faithful to the command but unsafe in the room, which is a different failure from hallucination and one that a literal command on a small rover will always allow.

Probe chains. The probes test what the pipeline does when one step cannot be carried out. In all 9 cloud runs and 12 of the 15 local runs, the planner dropped the unresolvable step without saying so and returned a seven-step plan for an eight-step command, so the

Table 6.4: Virtual chains by kind and depth. Each cell gives correct plans and completed missions out of the runs ("Corr., compl."), and the median planner latency in seconds ("Plan").

		Local (Gemma 4 E4B)		Cloud benchmark	
Kind	Depth	Corr., compl.	Plan (s)	Corr., compl.	Plan (s)
approach	1	5/5, 5/5	1.0	3/3, 3/3	1.2
approach	2	5/5, 5/5	1.2	3/3, 3/3	1.8
approach	3	5/5, 5/5	4.0	3/3, 3/3	1.8
approach	4	10/10, 10/10	4.2	6/6, 6/6	1.9
approach	6	10/10, 10/10	2.5	6/6, 6/6	2.3
approach	8	10/10, 10/10	2.7	6/6, 6/6	2.3
approach	10	10/10, 10/10	6.0	6/6, 6/6	2.3
approach	12	10/10, 10/10	4.7	6/6, 6/6	3.1
literal	2	5/5, 5/5	1.4	3/3, 3/3	5.2
literal	4	5/5, 5/5	1.9	3/3, 3/3	8.5
literal	6	5/5, 5/5	3.4	3/3, 3/3	6.1
literal	8	10/10, 10/10	3.1	6/6, 6/6	8.3
mixed	4	5/5, 5/5	2.0	3/3, 3/3	7.1
mixed	6	5/5, 5/5	5.6	3/3, 3/3	8.8
mixed	8	5/5, 0/5	7.5	3/3, 2/3	12.3

mission continued past the missing fridge or the unsupported "pick up the lamp". The remaining three local runs, all on the chain ending in the television, kept the step, and two of them halted on it, which is the intended behaviour. No unresolved target was logged for any of the nine cloud probes or the twelve silent local probes (`probe_logged` false), and one local run was counted as a hallucination. The silent dropping of steps is the most important failure in the virtual tests, because it produces a plan that looks valid.

Context window and the plan ceiling. Figure 6.1 shows the context sweep. With the configured `num_ctx` of 8192 the local planner produced correct plans for chains up to 36 instructions in all three runs, and at 40 instructions it produced a 34-step plan in all three runs, dropping the last six visits. The ceiling is therefore a plan-length limit of the model, and not a context limit, since it appears at 8192 where the plan uses only a fraction of the window.

Table 6.5: Correct plans out of three runs, by context setting and chain depth (Gemma 4 E4B).

`num_ctx`	Depth 4–32	36	40	Empty plans
8192	3/3 at every depth	3/3	0/3 (34 steps)	0
2048	3/3 at every depth	2/3	0/3	1 (depth 40)
1024	3/3 at every depth	2/3	1/3	1 (depth 40)

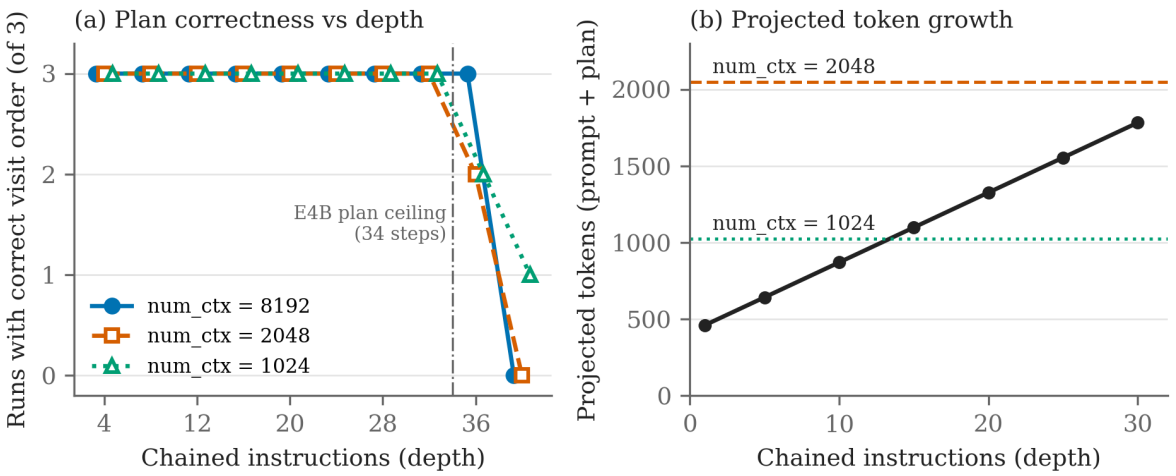


Figure 6.1: (a) Plans with the correct visit order out of three runs, against chain depth, at three context settings (points are offset horizontally for legibility). (b) Projected tokens per planning call against depth, measured to depth 30, with the 1024 and 2048 limits marked.

Measuring the planning prompt gave 460 tokens at depth 1 and 1 782 at depth 30 (about 46 tokens per added instruction), which is 22% of the configured 8192-token window. For a mission that includes replanning with three guidance turns, the largest prompt measured was 1 271 tokens at depth 30, and the largest revision prompt 1 579 tokens. Extrapolating the single-shot slope gives an overflow at about depth 170 for 8192 tokens and about depth 2 800 for the native 128K window, far beyond the 34-step ceiling. The same projection predicts overflow at about depth 13 for 1024 and about depth 36 for 2048 tokens. Table 6.5 does not follow this for 1024: plans stayed correct up to depth 32 at a setting where the projection says the prompt should not fit. The prediction at 2048 matches the observed drop at depth 36 only by coincidence, since the same drop occurs at 8192. One empty plan at depth 40 appeared at each reduced setting and none at 8192, which is suggestive but is one run each.

Candidate: I could not verify that the server enforced the reduced context. Run ollama ps during a sweep (or an overflow probe longer than 1024 tokens) and state the loaded context length here before using the 1024 and 2048 results as evidence either way.

6.5 Reference Comparison against Cloud Models

The cloud benchmark (gpt-5.6-luna through the OpenAI API) matched the local pipeline on plan correctness in every non-probe chain (63 of 63 against 105 of 105), and it behaved the same way on the probes, with the unresolvable step dropped silently in all nine runs. Across the non-probe chains the median planner latency was 3.1 s for the local model and 2.6 s for the cloud model, and the cloud model was slower on literal and mixed chains, where it needed 5 to 12 s against 1.4 to 7.5 s locally (Table 6.4). In these tests the cloud model had no advantage in correctness and none in latency, which is consistent with the project's premise that a small local model is enough for the plan-generation step. Two

limits apply. The cloud runs stop at depth 12 for named landmarks and did not include the depth-40 sweep, so the 34-step ceiling cannot be compared, and no cloud run was made on the physical rover.

6.6 Discussion

RQ1. On the physical rover, the local pipeline handled levels 0 to 2 without help in all ten runs, and levels 3 to 5 ended in a completed mission in 11 of 13 runs (nine directly and two after guidance), with one run halted after guidance at level 3 and one aborted at level 4. Its main weakness at level 4 was dropping the condition in two of seven plans, and its main weakness at level 3 was speech-to-text noise, which twice changed or lost the meaning of the command. Reliability at levels 3 to 5 is therefore supported by a handful of runs and should be reported as a demonstration of capability, not as a rate. The part of the design that earns its place is the guidance loop, which turned several runs that would have ended with "target not found" into completed missions.

RQ2. In the virtual tests no hallucinated step appeared in chains of up to 12 named landmarks or 8 literal instructions, and chains of up to 36 instructions stayed correct at the configured context. The first failure mode is not hallucination but truncation, at about 34 steps, followed by the silent dropping of steps that cannot be carried out. Neither failure is caught by the pipeline at plan time, which argues for a capability check on the plan before departure as the most useful next improvement.

Limitations. The evidence is one room, one rover, and three to seven runs per physical level. Level 5 is not yet adequately covered, the level 4 branches have not been scored against the true scene, and the vision-based distance estimate was not good enough to use. The virtual tests switch odometry drift off and use typed text, so they separate plan quality from perception and speech, and they are not directly comparable to the physical runs. The ultrasonic stop does not protect against low obstacles approached at an angle.

Chapter 7

Conclusion

7.1 Answers to the Research Questions

7.2 Contributions

7.3 Limitations

7.4 Recommendations for Future Work

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Appendix A

Appendix title goes here